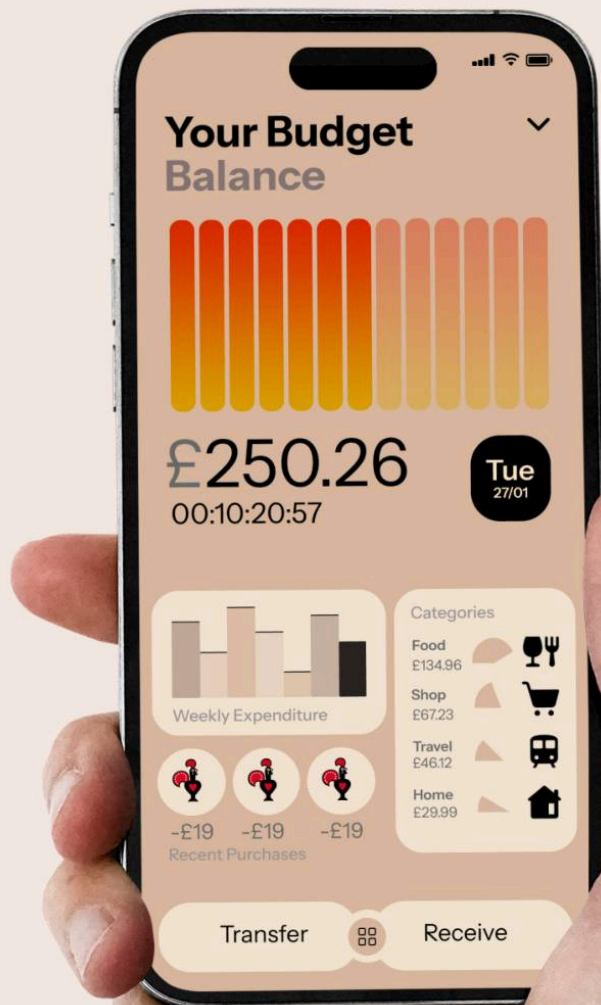


Cardo

Budgeting made easy



Business Report

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1 Executive Summary

Problem

Managing finances effectively remains a significant challenge for young professionals. Modern Neo Banks and traditional banking solutions attempt to address the situation by developing in-app budgeting. However, they fail to provide intuitive spending insights, lack personalisation and are tough to use. This leaves users struggling to make informed financial decisions, resulting in debt and minimal savings. To educate, nudge and aid them in cutting costs, whilst having a sustainable business model is a key opportunity that our solution Cardo can aim to address.

Solution

Cardo addresses this gap as a neo-bank offering a seamless financial management platform with integrated electrochromic smart cards and ML-driven budgeting insights. Our solution uses real-time spending categorisation, personalised feedback on inefficient spend patterns, and visual cues to empower users to control their finances effectively. To make this lucrative for businesses as well, spending data is essential to understanding general consumer demand, thereby highlighting how Cardo also bridges a mutually beneficial relationship between the user and business.

What makes Cardo unique is that it can act as a day-to-day expenditure and savings platform; a unique current account users can add on to their existing accounts as both our digital and physical design is different from current market.



Image 1.1 Cardo Mockup

Key Features

- *Electrochromic Card: Real-time colour changes reflect spending against set budgets.*
- *Machine Learning Insights: AI algorithms, including AWS Kinesis and ARIMA models for regression analysis to categorise spending and provide tailored financial suggestions.*
- *Data Security: AWS Key Management ensure end-to-end encryption and secure user data handling. Data is also stored anonymously via cloud databases.*

The process

The process to refine to reach this final concept, and breaking down how it can function as a business is described in the following diagram. It provides a distinct summary of how the business was formulated and forecasted for the next 3 years.



Table 1.2. Report Overview

Conclusion

Cardo combines cutting-edge technology, personalised financial tools, and an engaging user experience to redefine budgeting for modern users. With a clear growth strategy, diversified revenue model, and robust infrastructure, Cardo is well-positioned to scale and sustain as a business.

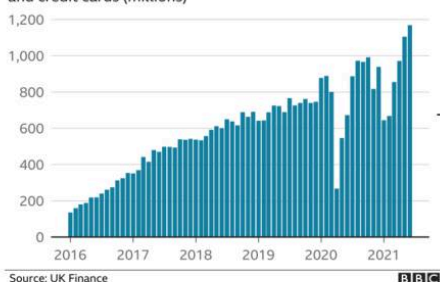
2 Market Analysis

2.1 Industry Scope

2.1.1 Key Market Trends

Growing use of contactless cards in the UK

Number of contactless purchases on debit and credit cards (millions)



Source: UK Finance

Figure 2.1.1: Contactless Card Growth (1)

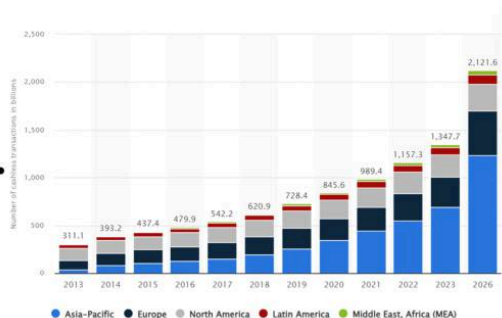


Figure 2.1.2: Global Contactless Transactions (2)

Neobanking Global Market Report 2024

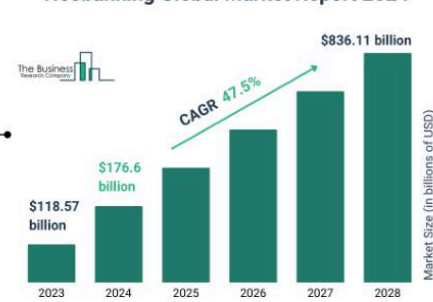


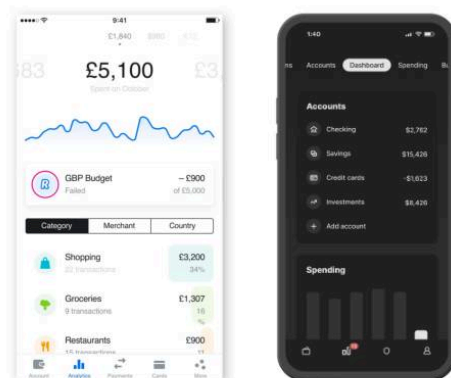
Figure 2.1.3: Neo bank Industry Growth (3)

3 key trends seen in the market is the growth contactless cards, the consequent growth in contactless transactions, and a bigger market share taken by **neobanks** as a result. Contactless cards, whether they be physical or via the phone (Apple Wallet, Google Pay, etc) hold a sizeable advantage to prior **chip-and-pin cards**, therefore being widely adopted in industry.

2.1.2 Growth in budgeting

Another trend seen is the growth in **budgeting platforms**. Apps like **Albert**, **Snoop** and banks like **Revolut** and **Monzo** are now implementing budgeting analytics. With the growth of the cost of living and wage stagnation for young adults, the need for **smart budgeting** is higher than ever.

Validated via a survey conducted in 2019 by CFP (Certified Financial Planner Board), which found that **'64 percent find it easy to overspend when using a credit card or mobile payment'** (4)



Revolut Budgeting

Albert (app)

Figure 2.1.4: E.g User Interfaces

Economic

Pricing the card competitively for different income brackets while covering production costs.

Growing interest in tools promoting budget management

Political

Must adhere to financial regulations and consumer protection laws in various regions.

Regulatory support for new fintech brands, e.g: open banking laws, relaxed licensing, etc

Legal

Securing patents for the physical counter technology to avoid legal disputes.

PESTLE Analysis

E

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E

Social

Rising demand for easier budgeting skills

Growth of a mobile-first culture in the new generation

Technological

Popular among younger generations seeking tech-savvy solutions and older users who prefer simplicity

Convenience of tech interventions, connected to smartphones, app ecosystem

Environmental

Using eco-friendly materials in the card's production to reduce environmental impact.

Need of adhering to data protection laws, privacy, etc

Figure 2.1.5: PESTLE Analysis

2.2 Competitor Landscape

2.2.1 Direct Competitors

Below highlights a direct competitor analysis to gauge why the key players in our industry excel in their respective fields, and where their fallbacks lie. This analysis helps us identify opportunities to differentiate our concept, address unmet customer needs, and strategically position ourselves in the market for maximum impact and success.

Table 2.2.1 Competitor Landscape

Competitor	 Emma	 YNAB	 Revolut	 Monzo
Pricing Model	Free basic; Emma Pro (£4.99/month or £29.99/year) for advanced analytics (5)	Subscription-only at £11.99/month or £84/year after 34-day free trial. (6)	Free basic; Premium (£6.99/month) and Metal (£12.99/month) tiers. (7)	Free basic; Monzo Plus (£5/month) and Monzo Premium (£15/month). (8)
Budgeting Features	Basic budgeting, categorized insights, and alerts; Pro includes multi-account tracking. (5)	Zero-based budgeting with goal tracking.	Budget categories, spending limits, and real-time analytics.	Detailed budget breakdown, spending insights, and real-time tracking with 'pots'.
User Experience	Simple dashboard with colorful insights; Pro adds visualization of spending	In-depth tools for zero-based budgeting on desktop and mobile. (6)	Modern interface, extensive tools, clear visuals.	User-friendly, clean design for budgeting and tracking.
Integration	Aggregates accounts into one dashboard. (5)	No bank connection; relies on manual input or file import.	Multi-bank, crypto, stocks, and multi-currency support.	Connects with Apple Pay, Google Pay, and UK banking features.
USPs	Financial "assistant" with subscription tracking and insights.	Rigorous zero-based budgeting; total spending control.	Low-cost international transfers, multi-currency, crypto and stocks.	Real-time spend notifications, savings Pots, and loans.

2.2.2 Indirect Competitors

Some indirect competitors include both major card manufacturers as well as banks/financial advisories:

- **Financial Advisors/CFAs:** Individuals or firms people reach out to for financial advice.
- **Investment Managers:** Companies that aid companies and individuals invest more efficiently through pre-existing portfolios, usually ranked from low, medium and high risk/returns.
- **Expense Tracking Apps:** Tools used to manage expenses in real-time. Can be used to catalogue
- **Affiliate/Discount platforms:** Companies like **Honey** and **Groupon** that provide live discounts, tips
- **Financial Literacy:** Platforms and online creators providing young people with financial advice
- **Price comparison apps:** These platforms provide live price comparison of different goods. Some examples include **Trolley**, which provide live comparisons of groceries, dry goods, etc.

2.3 SWOT Analysis

After determining the most important competitors in the market, we conducted a SWOT analysis of the whole market. This gives a thorough breakdown into gaps within the market, and recognising industry-wide trends as a result.

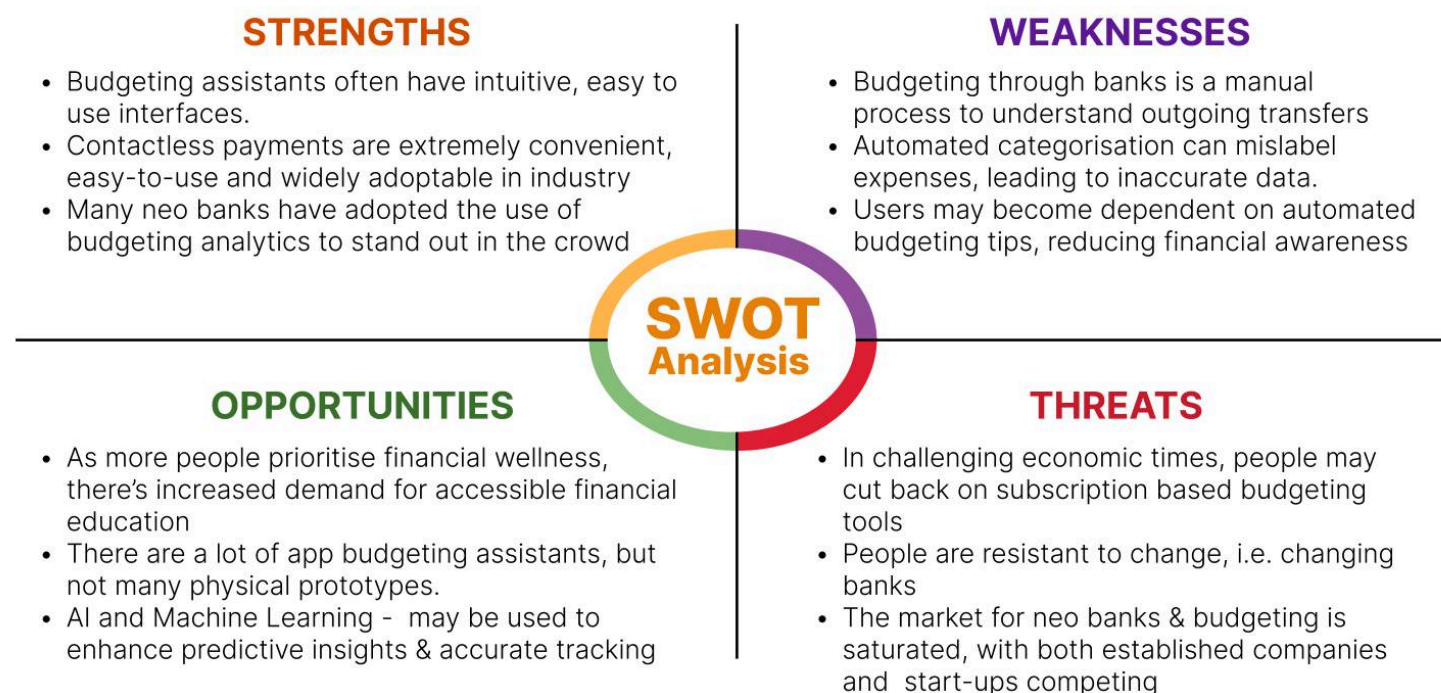


Figure 2.3.1: Market Analysis - SWOT Analysis

2.4 Key Competitor

2.4.1 Monzo

Through the competitor landscape and a SWOT analysis of the market, it was found that the key competitor in the current UK market for Cardo is **Monzo**. They are the most similar competitor to our value proposition of a neo bank built on a day-to-day budgeting; a solution that can complement other savings accounts well.



1. Offers popular budgeting features like "**Salary Sorter**" and "**Savings Pots**"
2. Support for joint accounts and innovative features like bill splitting
3. Automated saving tools like "**Round-Up Savings**" makes saving convenient

Biggest Threat

1. Monzo's **loan book income**, comprises of **25%** of their **£880 million revenue** in 2023 (4), benefiting from high interest rates in the short term. But faces risks from rising defaults and living costs in the UK (9).

How is Cardo unique?

Since neo banks like Monzo rely on income through interest payments and high volume interchange fees, there is an **opportunity** to **diversify income streams**. They also **offer manual budgeting**, a trend seen in all challenger banks in the market today, but allows room for **new disruption**.

Cardo stands out by acting as a **complementary, day-to-day** bank for **variable costs** in a month, week, even a day. Here are some key features that differ from current neo banks:

1. **Hybrid** of an **automated budgeting assistant** and **neo bank**; with a **tactile, hue-changing card**
2. Spending data analysed in real-time to provide **nudges, suggestions** to **cut costs** (unique to Cardo)
3. Our **business model** is **different**, as income is generated from both **B2B** and **B2C** methods:
 - a. **B2B**: advertisements on app, demographic spending data sold to businesses, partnerships
 - b. **B2C**: subscriptions, interchange fees via transactions,

2.5 Internal Risks

Table 2.5.1 Internal Risks

It is essential to conduct a risk analyses of the major internal & external risks threatening the business. After identifying them, it is also necessary to devise viable contingency plans to address the issues, and as a consequence calculate a resolved impact & new probability.

Risk	Impact	Probability	Contingency Plan	New Impact	New Probability
Relevance of ads and nudges	High	Medium	Avoid - Data-driven insights is key, Segment users by personal behaviours, Dynamic ad targeting to ensure it aligns with individual needs according to day, time, season, etc	Low	Low
Inability to rapidly manufacture electrochromic cards	High	High	Avoid - Use multiple suppliers, optimise through prototyping, launch pilot programs in limited markets to identify manufacturing issues, stockpile an inventory of cards Active Acceptance - Allocate 10-15% of card budget for expedited shipping, stagger deliveries	Medium	Medium
Privacy concerns for spending data	Very High	High	Contain - Store user data anonymously, stripping personally identifiable information (PII), Hire a cybersecurity firm every quarter to audit systems, Implement AES-256 encryption	Medium	Low
Scalability issues to address demand	High	Medium	Contain - Plan for future growth, invest early into outsourcing, and build a robust cloud infrastructure. Scale incrementally every month	Low	Low
Affiliate model reliance	Medium	Medium	Active Acceptance - Boost marketing & benefits for subscriptions, diversify income e.g: sell more aggregated data. Identify affiliate underperformance through regular checks	Low	Medium
Low interchange fee income	Medium	Medium	Avoid - Focus on high frequency, low-ticket transactions by advertising day-to-day functionality. Incentivize users with cashback	Low	Low

2.6 External Risks

Table 2.6.1 External Risks

When identifying external risks, it was essential to consider different aspects that could affect the business. For example, risks could be legal; data breaches, IP infringement, or market related; like brand loyalty and increased competition. Taking these into consideration is crucial.

Risk	Impact	Probability	Contingency Plan	New Impact	New Probability
IP infringement of electrochromic cards	High	Medium	Avoid - File patents for the electrochromic technology and any proprietary algorithms used. Ensure trademark and copyright protection. Lease to other banks if demand is present	Low	Low
Increased competition; other companies using more budgeting features	Very High	High	Active Acceptance - Dedicate resources to UX research to drive regular updates of budgeting features (bill negotiation, indirect nudges, etc). Emphasise USP of interaction of physical and digital sides to drive brand loyalty.	Low	Medium
Violation of data privacy regulations	High	Medium	Avoid - Implement an opt-in system where users must explicitly consent to data collection, Use multi-factor authentication (MFA) for users and employees.	Medium	Low
Customer loyalty to other banks	High	High	Active Acceptance - Conduct regular market analysis & highlight opportunities. Develop relationships with affiliates to provide unique deals, drive USP of day-to-day financial management; different from other banks	Medium	Medium
Sudden changes in consumer spending	Medium	Medium	Contain - Use streaming data frameworks like Kafka or AWS Kinesis to compare past habits with changes. Also use time-series models like ARIMA to adjust for expected fluctuations	Low	Low
Affiliate fraud or weak partners	Medium	Medium	Contain - Vet affiliates, monitor affiliate performance, trackKafka or AWS Kinesis clicks, conversions	Low	Low

3 Design Journey

This section below details the primary research from interviews and surveys, and secondary desk research conducted which influenced our concept ideation. Following an extensive market analysis a series of concepts were generated to selected and refined from user feedback to produce our final developed concept



3.1 Stage 1: Primary Research

In the initial phase we conducted a series of 18 in-person interviews to gauge an initial understanding of different peoples habits [1]. In addition to this we generated an online survey by which we gathered 57 responses from ages 18-70 to gain a holistic overview of potential users and identify patterns between demographics and their respective budgeting habits [2].

3.1.1 Interviews

“ I don't like checking my account because it feels **overwhelming**, but something subtle would be good. ”

“ I'd love something **simpler** than Excel; it just takes too much **time** to set everything up manually. ”

“ Budgeting apps feel too **complicated** for me, and I don't think I use half the features they offer. ”

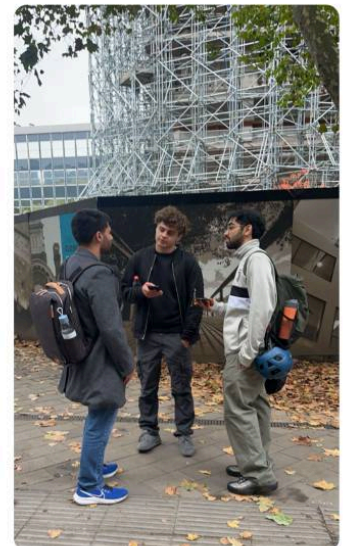


Figure 3.1.1: In Person User Interviews

3.1.2 Survey

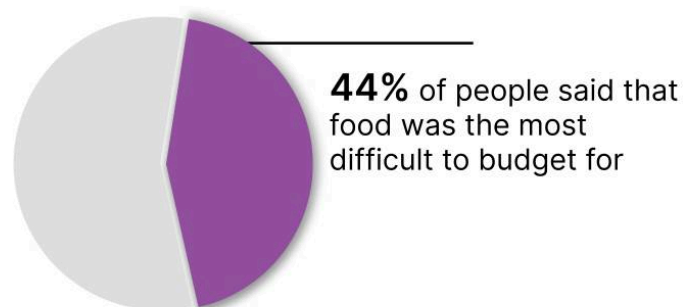
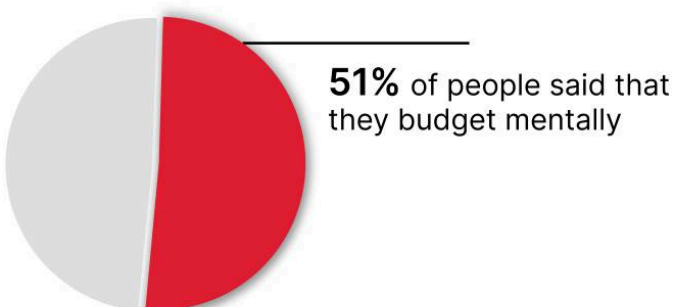
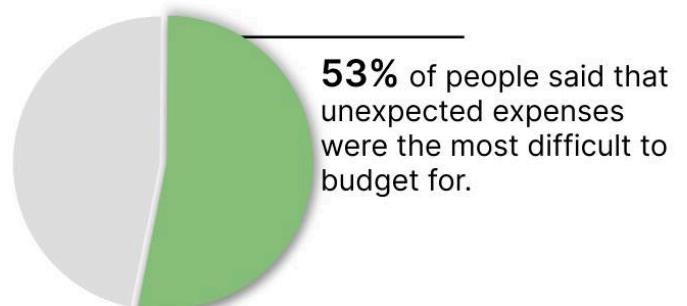
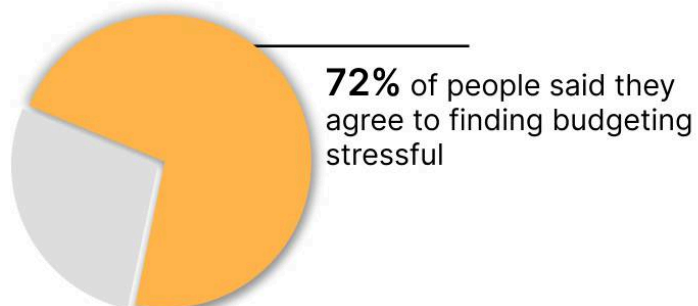
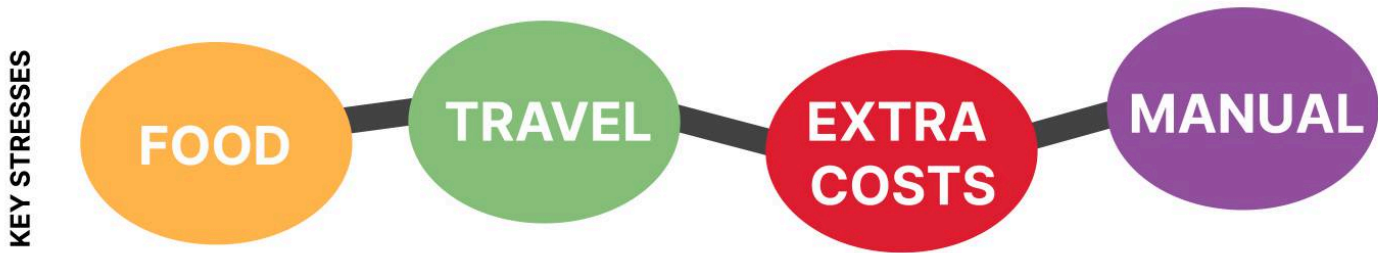


Figure 3.1.1: Pie Charts of Key Results

3.2 Stage 2: Initial Ideation

Based on the interviews and surveys we conducted, we gathered key insights from each participant, identifying recurring themes and challenges faced by respondents. These insights were then grouped into categories such as food, transport, subscriptions, and other common issues. Using this information, our team engaged in an ideation process to explore innovative solutions addressing these challenges. From this, we selected the five most promising ideas, which are detailed below.



3.2.1 Expense Forecasting App

The app uses machine learning to analyse patterns in variable expenses (groceries, entertainment, etc.) and provides predictive insights. It helps users anticipate fluctuations and better manage their finances.

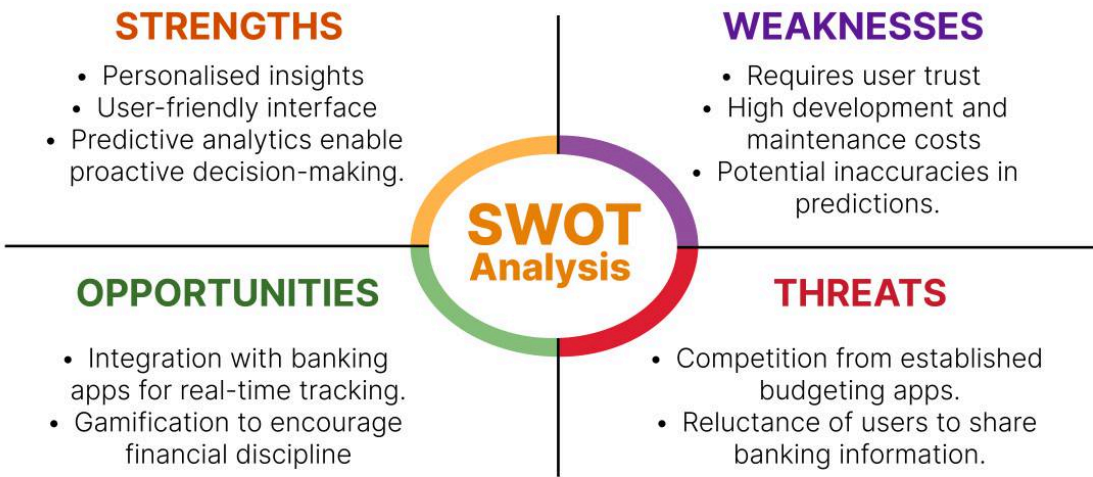


Figure 3.2.1.1: Concept 1 - SWOT Analysis



Figure 3.2.1.2: AI Generated Concept - Forecasting App

3.2.2 Haptic Feedback Keychain

A keychain attachment that buzzes when users approach their spending limit, creating a tangible and immediate budgeting reminder.

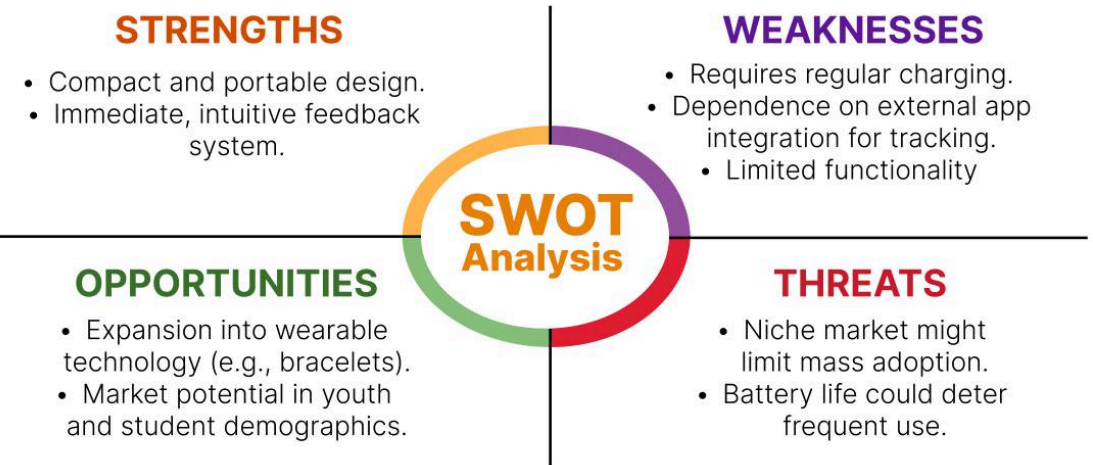


Figure 3.2.2.1: Concept 2 - SWOT Analysis



Figure 3.2.2.2: AI Generated Concept - Keychain

3.2.3 Home Voice Assistant

A voice assistant summarises daily purchases after contactless payments. It provides suggestions, predictive analytics, and helps users keep track of spending.

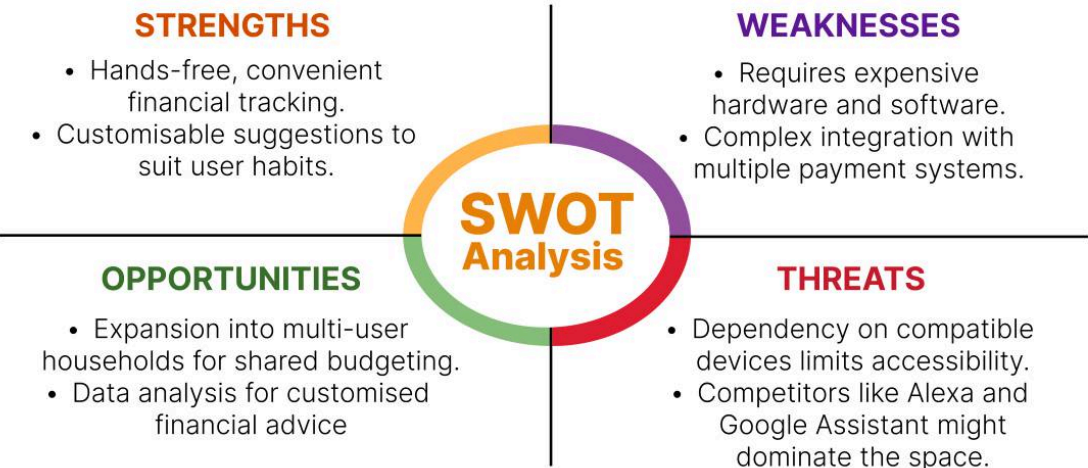


Figure 3.2.3.1: Concept 3 - SWOT Analysis



Figure 3.2.3.2: AI Generated Concept - Voice Assistant

3.2.4 LED Budgeting Card

A physical card equipped with an LED strip that visually tracks spending against a budget. The strip fills up like a scale as users spend, offering a clear visual representation of their limits.

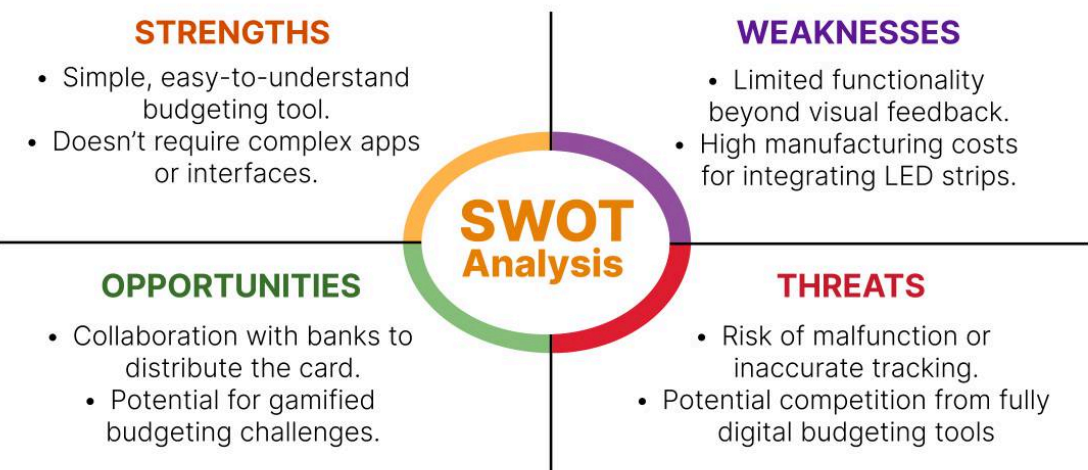


Figure 3.2.4.1: Concept 4 - SWOT Analysis



Figure 3.2.4.2: AI Generated Concept - LED Debit Card

3.2.5 Excel Assistant Plugin

A plugin that simplifies budgeting in Excel, making it more intuitive and accessible. Includes pre-set templates, automation, and interactive visualisation tools.

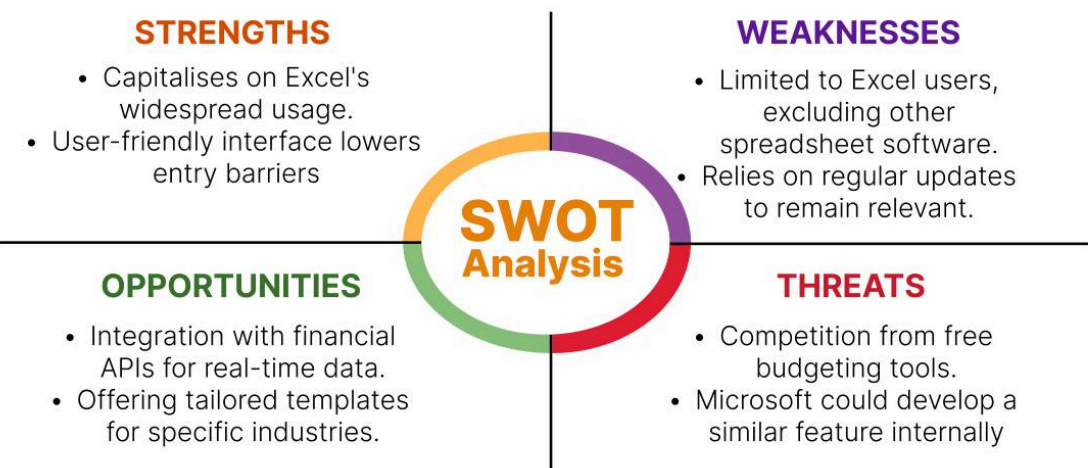


Figure 3.2.5.1: Concept 5 - SWOT Analysis



Figure 3.2.5.2: AI Generated Concept - LED Debit Card

3.3 Stage 3: Concept Selection

After conducting a SWOT on each idea, we decided to use a weighted decision matrix to show our findings

Table 3.3.1: Concept Evaluation		Expense Forecasting		Haptic Feedback Key		Voice Assistant		Budgeting Card		Excel Assistant Plug-in	
Criteria	Weight	Score	Total	Score	Total	Score	Total	Score	Total	Score	Total
Feasibility	3	4	12	3	9	3	9	5	15	2	6
Impact	3	4	12	3	9	3	9	5	15	2	6
Originality	2	3	6	4	8	2	4	5	10	3	6
Adoption	2	4	8	2	4	3	6	4	8	2	4
Cost	1	4	4	3	3	2	2	3	3	3	3
			42		33		30		51		25

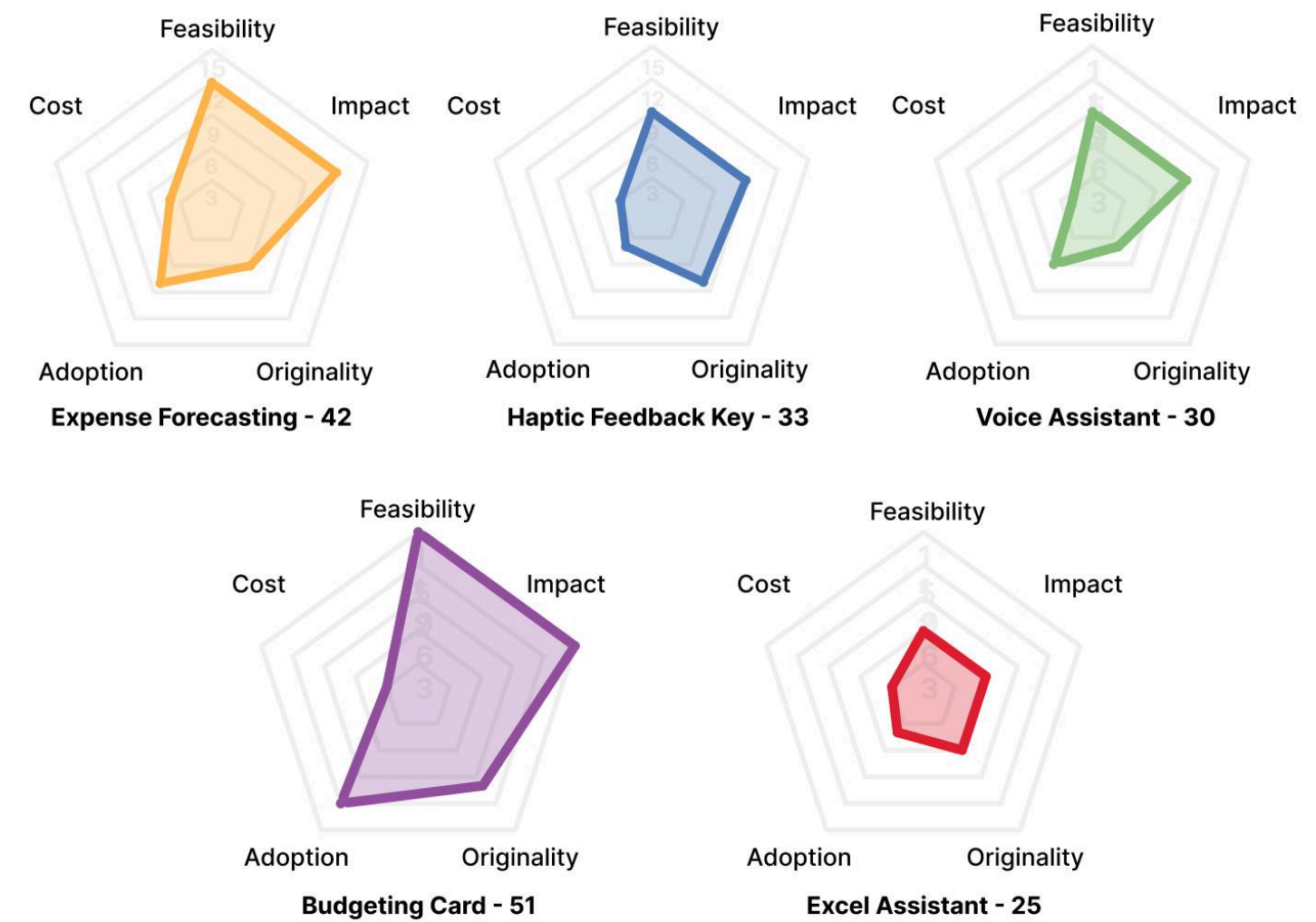


Figure 3.3: Radar Graphs of Our Concepts

After conducting a thorough concept evaluation of each idea, it was found that the top 2 ideas were the **budgeting card** and **expense forecasting**. To maximise the strengths of both; one leveraging the idea of a more tactile way of budgeting along with a quantitative method of forecasting money spent we found was necessary to address both needs for a consumer. A two-pronged solution we then visualised what this final concept may look like.

3.3.1 Initial Concept Mockups

Key Features

01. Day-by-day spending data

Users can keep a day-by-day log of their financial spending. Each day would show if you are over/under budget,

02. Transactions and Budget Page

Transactions shown as widgets on page, limit clearly shown in middle. Date, time left before budgeting finishes, and the background corresponds to money spent.

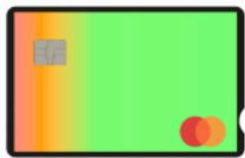
03. Detailed Insights on spending

Following short summaries on purchases, this page will provide the user with insights on their spending patterns. Would show their growth, days they generally lack on, etc.

App-side UI



Figure 3.3.1: Initial UI Design



Under budget



Figure 3.3.2: Initial Card Design



Above budget

04. Electrochromic Hue Changing

After conducting some research, we felt it could be beneficial to have a visual cue for spending. To apply this, this can be achieved through the use of electrochromic means; helping us be different from conventional cards

What is 'electrochromic' technology?

Electrochromic technology functions by using a low-voltage electrical current to cause ions to move between layers, altering the material's tint/colour [3]. It is currently observed in smart glass glazing, mirrors, cars, etc.

They typically have a multi-layer design, having the following:

- **Substrate:** A flexible base material (e.g. plastic) that provides physical support for the layers
- **Electrolyte:** Enables ion flow between the ink and electrode, causing the colour change
- **Spacer:** A layer that creates separation between the conductive layer and electrolyte
- **Electrochromic Ink:** This active material changes colour or opacity in response to electrical voltage.
- **Conductive Layer:** A thin layer, usually made of Indium Tin Oxide (ITO), that allows electric current to flow through the device. [4]

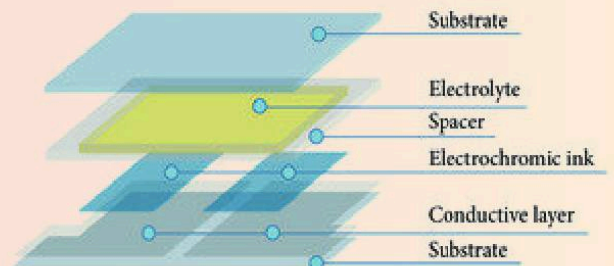


Figure 3.3.3: Electrochromic Technology Teardown [4]

Other details

Based on the feedback gained and our initial concept mock-ups, these are key details of our solution:

- **Improved Connectivity:** Incorporate secure **micro-BLE chip** to enable offline functionality and ensure seamless and reliable real-time data transfer between the card and app.
- **Optimised Power Solution:** Using an energy-efficient electrochromic display allows users to check their budget on the go. Powered by card machine to change on each transaction and hold colour until the next transaction from a card machine eliminates the need for charging.

3.4 Stage 4 : Validation

After final concept development, in order to make sure our concept aligned with stakeholder needs we created a survey on Qualtrics to validate our concept with our

For our validation survey we focused on responses from our primary user demographic, young professionals.[5] We took a qualitative approach to the survey to identify users thoughts about the user interface, card and overall concept as a whole. Using all 26 responses, to the questions in our survey, we identified the helpfulness of our concept and we performed a thematic analysis to Identify common themes form the responses.

3.4.1 Key Figures

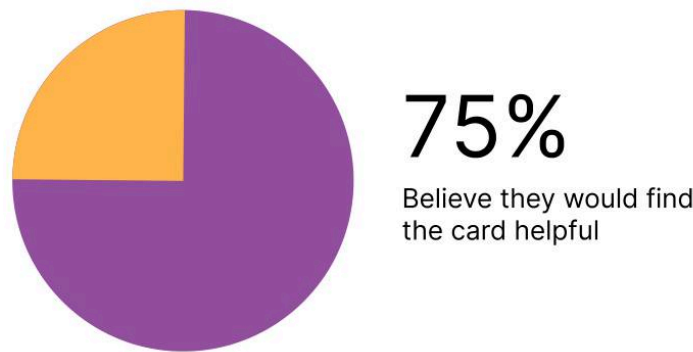


Figure 3.4.1: Pie Chart of Key Results

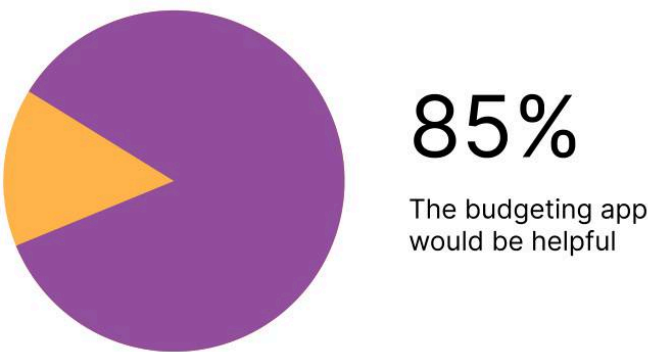


Figure 3.4.2: Pie Chart of Key Results

Through thematic analysis, we identified key insights highlighting specific areas where our final concept could be enhanced to address unmet needs effectively. These areas of improvement are detailed below:

3.4.2 Refinement

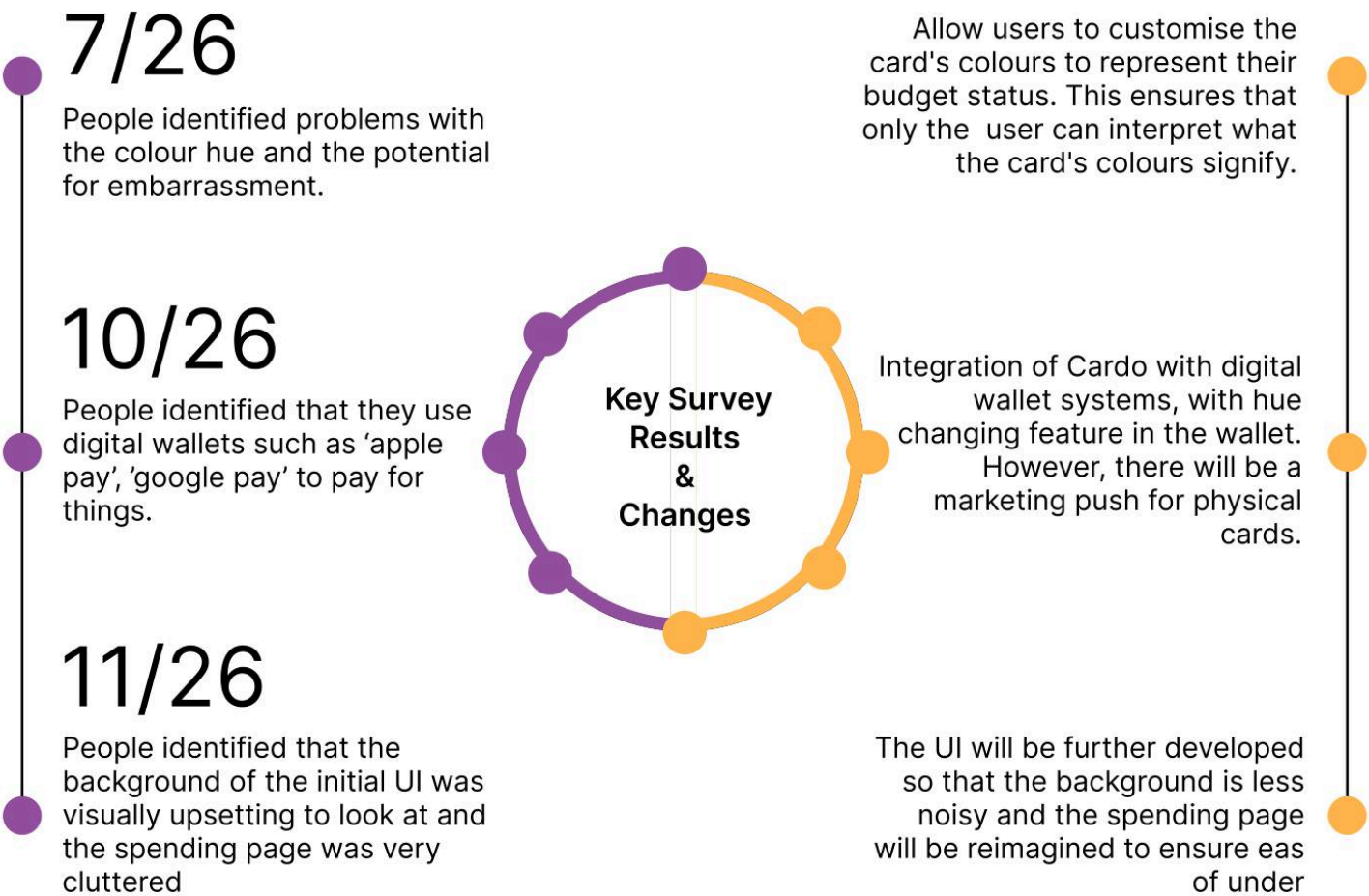


Figure 3.4.2.1: Overview of Key Qualitative Points

After our stakeholder survey we wanted to verify the feasibility of our idea by talking to experts. We reached out to people who work in similar fields of business in which we are trying to break into. We also had contact with a potential user getting a more in depth understanding on a customers perspective.

3.4.3 Stakeholder Validation

Young Professional at Unilever

Key Insights



Anda Krasniqi, 23

"When I first received my first paycheque, I didn't know what to do with that money. I didn't know how to start a pension or save money."

She spoke about how when she first started earning money she lacked the financial literacy to know what to do with it. She spoke about being "unaware of the repercussions of opting out of pensions or saving money for the long term".

"Most young people have got multiple cards, like a Monzo or a Revolut card, because of the travelling perks as well as their standard bank account."

She also validated that young professionals like her and her friends have "less loyalty to banks now and would be willing to switch, especially if they offer something unique".

Product Owner at Revolut

Key Insights



Louis Cutner, 24

"Current neo banks don't provide smart budgeting insights. Nudges through Cardo might be useful to incite behavioural change"

He spoke of the lack of smart budgeting insights in current apps, including Revolut. He is a frequent user of the budgeting features, but says it is inconvenient as it doesn't take into account context, unexpected purchases, etc.

"Budgeting on Revolut is too manual, requires a lot of my oversight. Breakdowns aren't detailed"

His money is divided between a current account and savings/rent. Budgeting is manual, as **Revolut** segments spending but lacks distinctions—e.g: "it doesn't separate groceries into supplies and ready-made food, making it hard to track expenses accurately"

"Think of partnerships with banks. They may be interested"

Leasing technology to other banks may be a good way of entering the market. It is a unique point that could make Cardo stand out.

Ex Property Analyst at Morgan Stanley

Key Insights



Sebastian Isola, 32

"There's very few current accounts really that offer interest"

From a business standpoint, this model enables us to achieve **higher revenue**. By earning interest on customer deposits without having to pay out interest to customers, our net interest revenue can exceed that of a traditional bank.

"They were basically a front for another bank"

Nonbanking more than often takes the name "white-label" [00000000] banking, this is because acquiring a banking license is tough so new banks like Revolut or Monzo often act as fronts for existing licensed banks.

3.5 Stage 5 : Cardo UI & Card (MVP)

Process is **step-by step**. Through the 3 pages, users can see their:

1. Budget
2. Weekly breakdown
3. Day-by-day analysis

UI was made less 'clunky' and more 'clear' as indicated through our **validation research**.

Inefficiencies break down sections that add upto wasteful spending. Gives cumulative insights & data for each day of the week, e.g: last 10 Mondays.

3. Inefficiencies Page

Shows a breakdown of inefficient spending and potential savings by day

2. Net Savings Page

Shows net savings for the week and percentage of potential waste.

1. Your Budget Page

Shows balance and time period of current budget. Gives other insights from day too.

Gradient Display

Visualises the budget left along the bars highlighted. In this case, the user is halfway through their weekly budget

Recent Purchases

Shows the most recent purchases made through Cardo. Visualises what company too.

Spending Categories

Breaks down weekly payments in separate categories. Can be expanded to give detailed breakdowns of each section.

Affiliate Ads based

on smart insights generated from a user's spending patterns



Personalisation of Colours

Users can choose personal colour schemes to indicate budgeting and spending. This can vary and be programmed via the MCU (Microcontroller Unit) in the card. This will directly relate to the spending shown on app

3.5.1 System Diagram

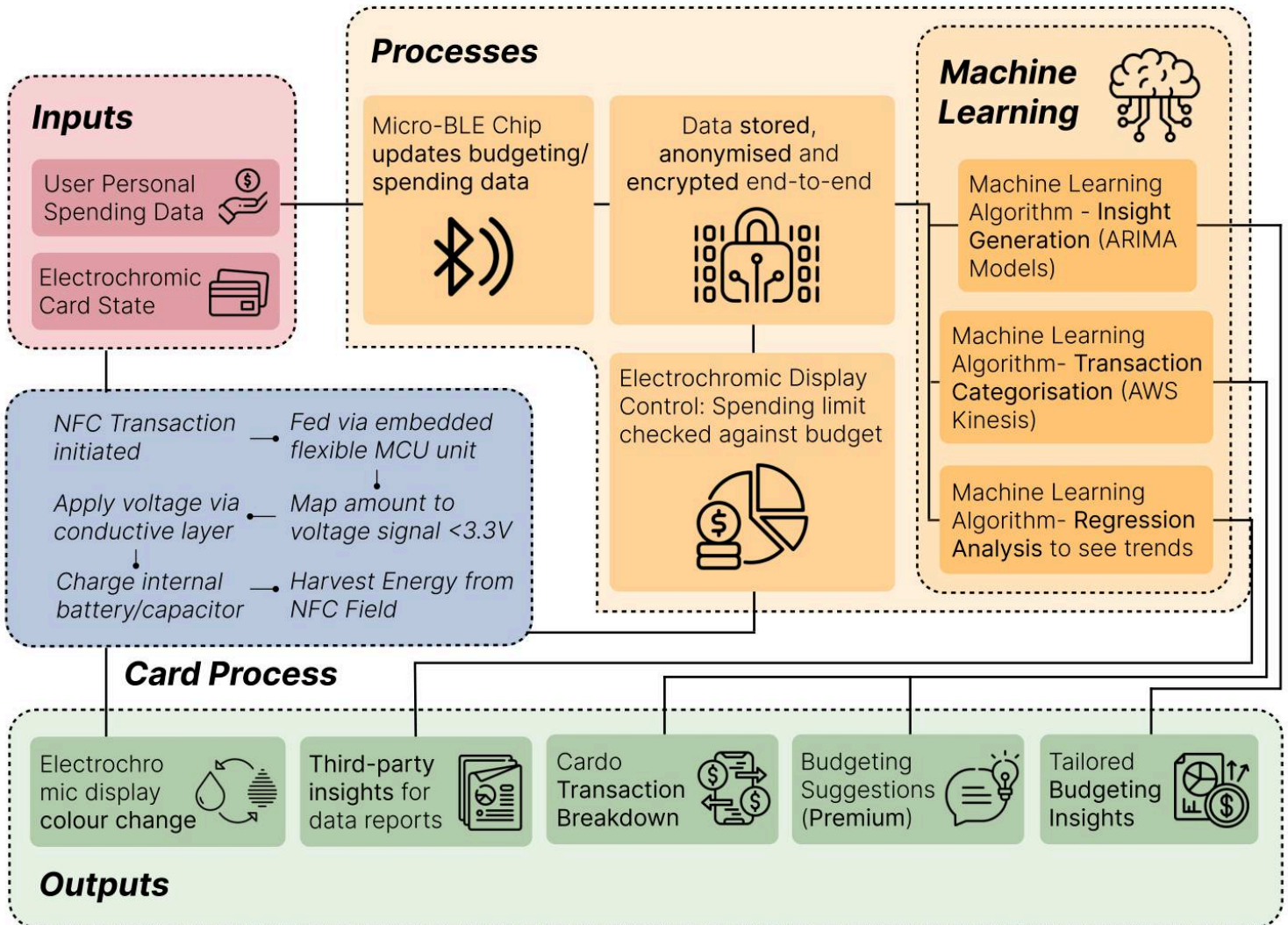


Figure 3.5.1.1: System Diagram

3.5.2 System Diagram Explained

The system diagram showcases the inputs the different processes carried out to then produce deliverables for our different customers so personalised insights for individual users, while for businesses reports on consumer spending trends.

User Spending Data: Transaction data, Spending patterns and budget inputs set by the user.

Electrochromic Card State: The cards current visual state, represented by its colour, reflecting user's spending status.

Micro-BLE Chip Updates Budgeting/Spending Data: A micro-bluetooth module imbedded in the card will transmit transaction data to your phone to update your transaction and budgeting history.

Data Stored, Anonymised and Encrypted end-to-end: All users's personal identifiable information (PII) is stripped and data is encrypted to prevent malicious data extraction.

ML Algorithm - Insight Generation (ARIMA): Time series forecasting to predict spending patterns.

ML Algorithm - Transaction Categorisation (AWS Kinesis): Real-time data platform to process transaction data and provide live categorisation and analysis.

ML Algorithm - Regression Analysis: Trends in spending data can be identified for users/businesses.

Electrochromic Display Colour Change: Visual indication of budget compliance or overspending.

Third-Party Insights For Data Reports: Anonymous aggregated spending data sold to businesses to gain market insights.

Cardo Transaction Breakdown: Detailed insights into user transactions displayed in the app.

Budgeting Suggestions (Premium): Personalised advice to optimise budget and spending

Tailored Budgeting Insights: Customised analysis based on individual financial behaviour to support better financial decision making.

4 Value Proposition

4.1 The Problem

Managing personal finances is a significant challenge for many, with budgeting often perceived as tedious, overly restrictive, or unnecessarily complex. The lack of effective tools to monitor spending in real-time leads to poor financial decisions, increased debt, and heightened stress.

4.2 Key Issues Identified

Research from the Financial Conduct Authority (FCA) reveals that **39%** of adults in the UK find managing money stressful [1], with many avoiding budgeting entirely due to perceived complexity or a dislike of numbers.

In 2024, the Office for National Statistics (ONS) reported that **68%** of households in the UK felt their cost of living had risen significantly in the past six months [2], leaving little disposable income to manage.

The Financial Lives 2022 survey by the FCA found that **24%** of UK adults had low financial resilience [3], indicating a vulnerability to financial shocks due to factors like insufficient savings or high debt burdens.

4.3 Our Value

01. **Real-Time Financial Awareness:** The electrochromic display instantly alerts users of their financial standing without needing to open an app or log into online banking. This empowers users to stay within budget effortlessly.
02. **Customisable Spending Controls:** Users can set personal budgets in the app (daily, weekly, or monthly). The card's visual alerts are directly synced, ensuring it aligns with each user's unique financial goals.
03. **Educational and Engaging:** The app leverages data from the card to create personalised insights, reports, and tips. Users can track patterns, learn about their habits, and improve financial literacy over time.



4.3.1 Novelty and Uniqueness:

- Real-Time Spending Feedback: Colour changing electrochromic technology alerts users as they near or exceed their budgets.
- Customisable Visual Alerts: Users set limits, and the card provides intuitive, actionable feedback.
- Integrated Ecosystem: Paired with an app, it offers real-time insights and smarter budgeting tools.



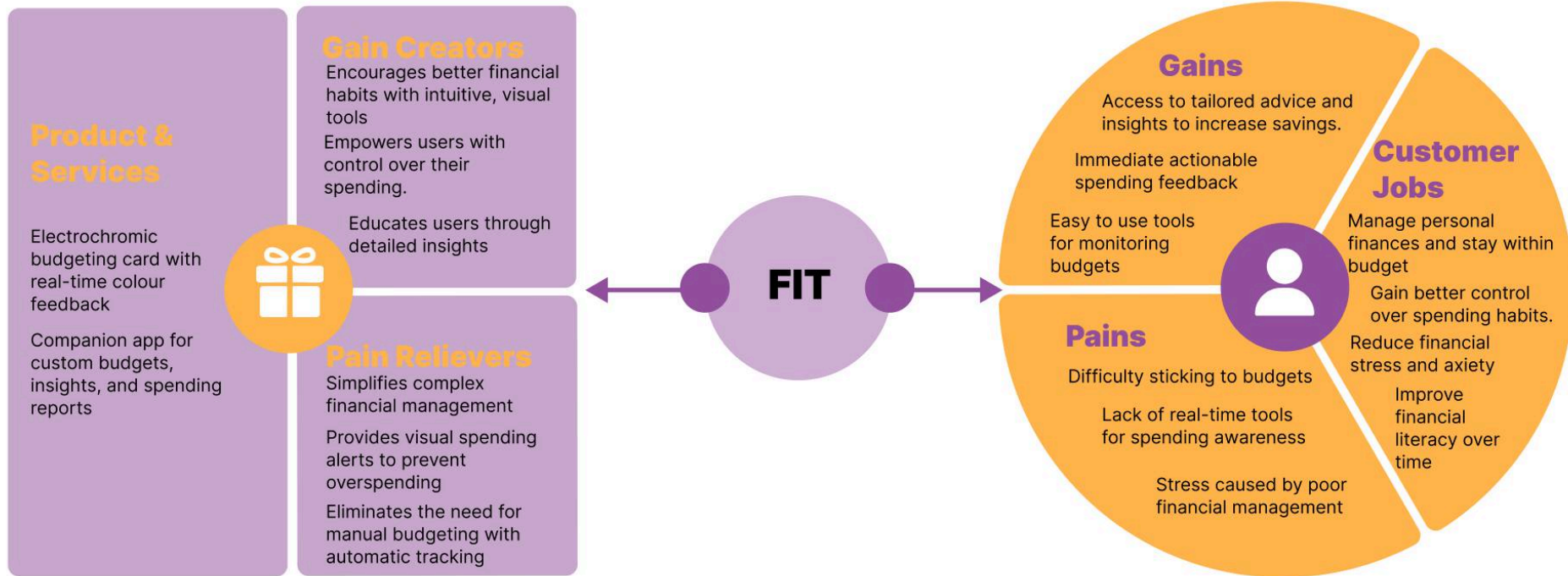
4.3.2 Competitive Advantage:

- Patent Potential: Electrochromism ensures a unique, protected solution with a competitive edge.
- Scalable Innovation: Adaptable for diverse financial products, expanding its market impact.

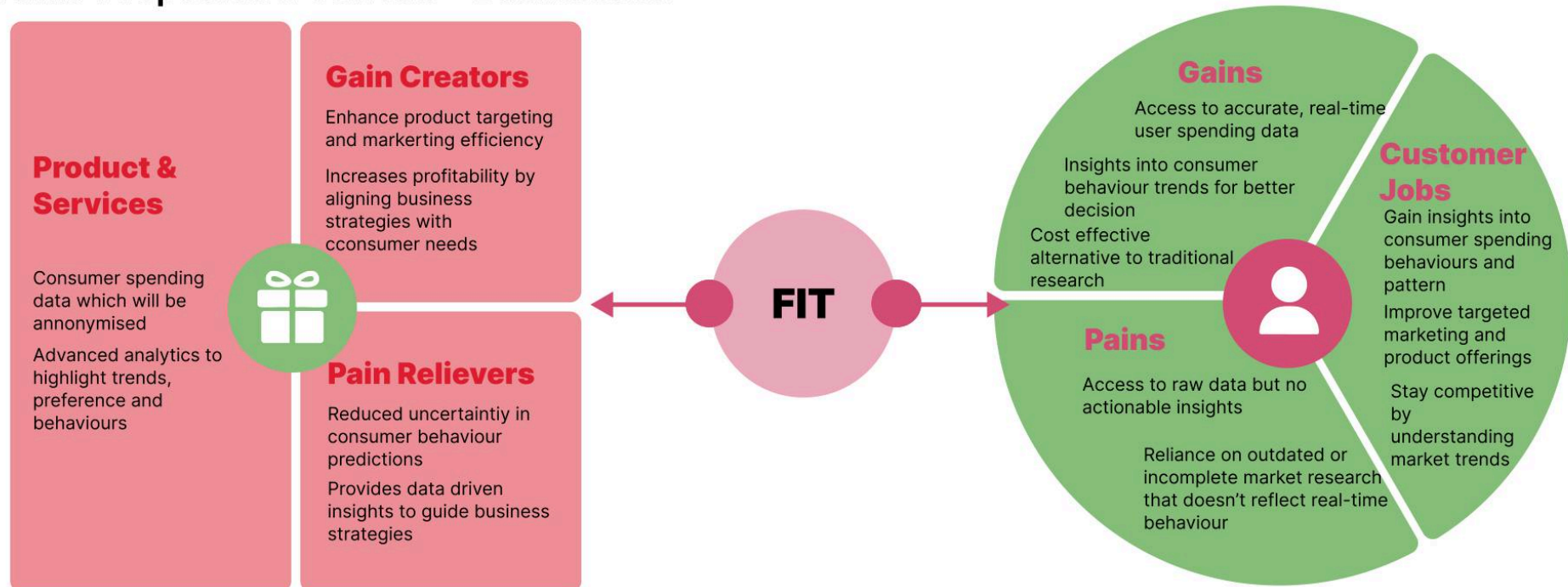
4.4 Value Proposition Canvas

A value proposition canvas was used in order to showcase how our concept delivers value not only to users but to businesses as well. Although our primary focus is on empowering consumers to become more financially literate and gain a better control over their spending, we also provide businesses with significant benefits, for example consumer spending trends. Given these distinct target audiences two separate value proposition canvases were created.

Value Proposition Canvas - Consumers



Value Proposition Canvas - Businesses



5 Market opportunity calculations

To thoroughly assess the market potential of the proposed digital reading platform, a comprehensive approach has been employed, incorporating three distinct methodologies: **top-down** analysis, **bottom-up** analysis, and **value theory** evaluation. Each method offers unique advantages and areas of focus.

5.1 Top Down Approach

The top-down approach adopts a macro-level perspective, utilizing **data** and overarching **market trends** to estimate the **potential market size**.

Total Adressable Market (TAM)

To estimate the TAM of the global neobanking sector, two methods were used.

Over 4.2 billion people in the world today have a digital banking app[1]. 250.74 million or 16.8% of these people have neobanking services.[2]. As a fraction of the retail digital banking market this represents **£350B**. [3]

The estimated Total Adressable Market (**TAM**) for the neobank sector in 2024 is **£100.6B** [4].

Therefore our TAM can be as small as **£100.6B** or as large as **£350B**.

£350B

£100B

T
A
M

Servicable Available Market (SAM)

To estimate the SAM we need to consider our target market. We found there to be **15 million young professionals** in the **UK**

In terms of revenue, UK accounted for 8.4% of the global neobanking market in 2022.[5] Young professionals in the UK are estimated to contribute 32% to this market. [6]

Applying this to our TAM we can estimate our SAM to be as large as **£9.4B** and as small as **£2.7B**.

£9.4B

£2.7B

S
A
M

Servicable Obtainable Market (SOM)

In order to calculate the **SOM** a sales channel was identified and then weighted based on primary research

In the uk people switching banks can be a great entry point for our product service, in the uk **1.38 million** people or **3%** of the SAM **switched banks** in **2023** [7].

From a survey, we found that **75% of people** would find our solution useful. This would account for **2.25%** of the **SAM**.

Applying this to our SAM we can estimate our SOM to be as large as **£211M** and as small as **£60.7M**.

£211M

£60M

S
O
M

3 YEAR **SOM** **£214M**

Figure 5.1 : Top Down Approach

with a **CARG** of 52.9 % [4]

5.2 Bottom up approach

The bottom-up approach builds a detailed market outlook by analyzing user segments, integrating insights from top-down competitive and market analysis. Assumptions are clearly stated, with further details in the appendix.

ADRESSABLE MARKET

Table 5.2.1 : Addressable market

Age [8]	16-35	36-60	60 +
Have banking [8]	90%	82%	44%
Budgeting issues [9]	39%	39%	39%
Addressable users [6]	5.3M	3.5M	2.53M

OBTAINABLE MARKET

Table 5.2.2 : Obtainable market

Switched banks [7]	3%	3%	3%
Obtainable users	159k	105k	76k

ARPU

Table 5.2.3 : Average Revenue Per User (ARPU)

Advertisement	19.44	19.44	19.44
Interest accrual	19.5	27.25	35
Subscription	9.6	9.6	9.6
Data Insights	31	31	31
Interchange fees	0.78	1.09	1.44
ARPU	5.3M	3.5M	2.53M

Tables 5.2.1, 5.2.2, 5.2.3 - Please check supplementary pages for detailed assumptions and calculations

The bottom-up approach yielded a serviceable obtainable market (**SOM**) of **£29M** consisting of **340,000 users**.

3 YEAR **SOM** **£103M**

with a **CARG** of 52.9 % [4]

5.3 Value Theory Approach

Impulse purchases accounted for **£416** per person in the uk in 2016 [10]. If our product service solution saves **50%** of this money each year for our **obtainable market**, this would bring a value to the consumer of **£70M**.

3 YEAR **SOM** **£250M**

with a **CARG** of 52.9 % [4]

5.4 Assumptions and sensitivity

Market Scope Assumptions

- 1 A compound annual growth rate (CARG) of 52.9% is assumed [4]
- 2 We aim to obtain 1.5 to 3% of our addressable users. [7]
- 3 People will use Cardo as an additional current account

Financial Assumptions

- 4 An interest rate that varies between 0.5-10%
- 5 Interchange fees between 0.1 and 0.3% [11]
- 6 'white label' [12,13] agreement, has a commission on interest and interchange fees of 30 - 40%
- 7 'white label' [12,13] agreement, or in the best case scenario a banking license

Operational Assumptions

- 8 Customer deposits average 491 per user
- 9 Data insights generated from our users provide value to other businesses
- 10 Advertisement generates income from impressions not clicks.

Figure 5.4.1 : Assumptions and Sensitivity

Sensitivity analysis

Due to the detailed nature of the bottom up approach it was selected to conduct a in depth sensitivity analysis.

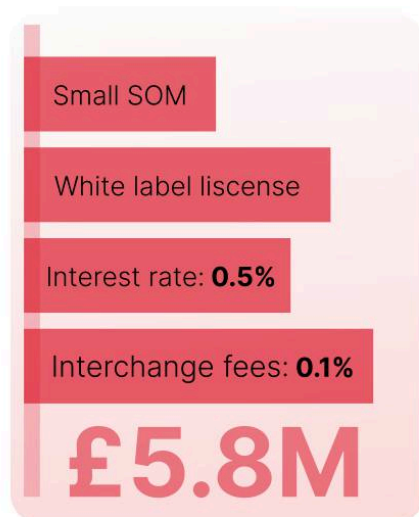


Figure 2.3.1 : Worst case scenario, calculations available on the appendix.



Figure 2.3.2 : Expected case scenario

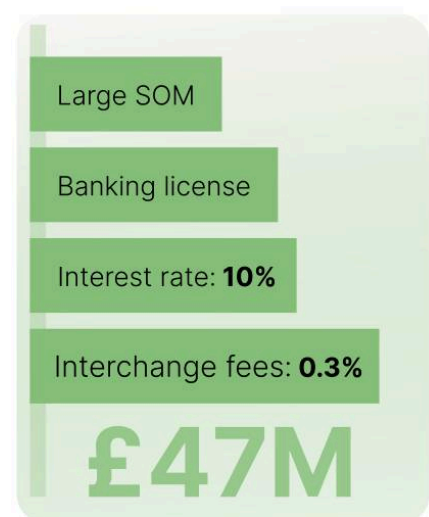


Figure 2.3.3 : Best case scenario

Biases and limitations

- **Our survey** was used to narrow down our addressable market, any biases in the survey can have a great influence on the potential obtainable market.
- **The obtainable market** was greatly narrowed down by assuming we can only capture those who are in search for a new banking app.
- **The market is currently limited to the UK**, representing small fraction of the Total Addressable Market (TAM). However, there is promising potential for expansion beyond the UK, which could result in market sizes significantly exceeding current estimates.

6 Business Model

Our Business Model Canvas shows how we build a platform that delivers real value to consumers and businesses. It breaks down the key elements of how we operate. Our model's heart is empowering our consumers with tools to improve their financial awareness and providing businesses with the insights they need to grow. We create a seamless and impactful solution for everyone involved with features like our electrochromic budgeting card, tailored spending insights, and predictive analytics. This canvas represents our roadmap for scaling Cardo and making a real difference in financial technology.

6.1 Business Model Canvas

Key Partners	Value Propositions		Customer Segments
Bank: - White label neobank People: - Payment Network : Mastercard - Software engineer: integration of machine learning and algorithm for budgeting - Financial advisors and Experts Infrastructure partners: - Data Centres - Electromechanical Card Manufacturer - Digital Transaction processor: Stripe	For Consumer: - Electrochromic budgeting card gives real-time colour feedback on budget - Companion app for hassle free budget plan, insights, and spending reports. - Increasing Financial awareness - Feedback on spending patterns	For Business: - Consumer spending insights: provides data driven insights to develop new strategies - Predictive analytics to highlight trends, preference and behaviours	Primary: Consumers- young professionals (20-35) Businesses -SME , e-commerce, retail Secondary: General consumer: 35 +
Key Resources	Key Activities	Customer Relationships	Channels
Physical: electrochromic card Human: Development team(app maintenance, card technology, data security), Sales and marketing team, Financial advisors and experts Intellectual: Patent protected electrochromatic card	- Software development team - Machine learning algorithm optimisation - Increase Awareness towards budgeting - Financial literacy - Banking license - Visual indicator card with electrochromic display colour change	Tailormade personalised insights: based on individual spending habits. Customer support: 24/7 online banking support B2B support: helping businesses utilise Cardos insights for better targetting and product development	For B2C: Community engagement: webinars Social media & influencer marketing Tube ads Organic channels: word of mouth, referral links For B2B: Trade shows: business engagement
Cost Structure		Revenue Streams	
Set up cost: - Patent Filings (IP) - White label banking services - FCA application fees - Compliance audits Electrochromic card manufacturing Back end services: data centres Marketing costs Employees wages Office costs/ rent and software pricing		Businesses: Interest accrual revenue Advertisement revenue (affiliate marketing) Aggregated Demographic Spending Data Interchange revenue Fee (IFR) Consumers: Card sale revenue Subscription revenue(Freemium model)	

6.2 Customer Segments

Understanding our customer segments is essential. It allows us at Cardo to strategise targeted marketing strategies, learn from our users and identify how Cardo can benefit them. Our customer segments are split into two primary customer segments. The 1st is our consumers, young professionals, our target users for the app, and 2nd is businesses such as retailers, e-commerce and small to medium enterprises. Our customer segment is categorised on pain points based on user research and how they align with Cardo.

6.2.1 Consumer

Young Professionals (20-35)

Pain Points

Business alignment

- Are receiving their first full pay checks and do not know how to spend their money.
- Find existing tools like excel too complex and too much hassle
- Improve financial literacy over time..
- Uncontrolled spending habits
- Unaware of money saving strategies and schemes available.
- Limited financial literacy

- Electrochromic card changes colour based on budget to physically indicate how well your budgeting is going.
- Cardo's hassle free budget tracking system helps users stay within budget.
- Automated budget helps users fix inefficiencies with targeted ads and suggestions, e.g., *linking a railcard to an Oyster*.
- Insight and report to educate users on better financial habits.

Figure 6.2.1 Consumer (20-35)

6.2.2 Business

Retailers/Superstore

Pain Points

E-commerce

Pain Points

Small Medium

Enterprises (SME)

Pain Points

- Difficult to understand demographical customer spending habits.
 - Struggle to target promotions effectively.
- Collecting market data to promote sales is expensive

- High cart abandonment rates from customer uncertainty about spending.
- Limited customer insights for personalising offers.
- Trouble competing with larger platforms.

- Limited customer data for tailoring products and services.
- Hard to compete with larger companies on advertising.
- Lack of tools for engaging with customers.

Business alignment

Business alignment

Business alignment

- Cardo provides spending insights through aggregated data.
- Better targeting of deals to customers and marketing strategies.

- Data provided will help e-commerce websites target customers more effectively.
- Products can be marketed in-app.

- Data on spending habits for more effective marketing.
- Support better product alignment with customer needs.

Figure 6.2.2 Business

6.2.3 Secondary

General Consumer (35+)

Pain Points

Business Alignment

- Find existing tools like excel complex
- Transition from card to contactless has caused more impulse spending.

- Cardo's simple design makes budgeting easy for all age groups.
- Provides real-time alerts to help users feel in control of their spending.

Figure 6.2.3 Consumer (35+)

We have both a business-to-business model and a business-to-consumer model. Both customer profiles are vital to Cardo. However, young professionals, our primary users, are essential for Cardo's existence, as their engagement drives the platform's success. Businesses will generate the most money for our revenue, but for this, we need consumers who are pivotal to Cardo's success.

6.3 Revenue Generation Strategy

Our Revenue Generation Strategy outlines how Cardo creates sustainable and scalable income streams by addressing the needs of both consumers and businesses. With a balanced approach to monetisation, we ensure that our users benefit from valuable tools for financial management, while businesses gain actionable insights and opportunities to connect with their audience. This strategy leverages multiple revenue models, each considered and tailored to maximise growth and profitability while aligning with Cardo's core mission of empowering financial literacy.

6.3.1 Business to consumer

01 Subscription revenue(Freemium model)

Cardo follows a freemium model, premium users pay £5 a month for tailor-made financial advice, money saving tactics and to turn off advertisements.

02 Card sale revenue

Cardo card is sold at a unit price of £10 per card.

6.3.2 Business to Business

03 Aggregated Demographic Spending Data

Anonymised data is collected from purchasing patterns behaviours and sold to businesses, i.e retailers, SMEs, e-commerce. Sold in reports of 20,000 users.

04 Interchange revenue Fee (IFR)

0.2% interchange fees generate revenue from every purchase. For every transaction there is a fee charged to the merchant this is a sum of the interchange fee (our cut), network provider fee(mastercard) and the acquirer fee (the card device).

05 Advertisement revenue (affiliate marketing)

Businesses can advertise within the cardo mobile banking app, this would come up in money saving insights. The revenue is based on a price per click model each click £19.44 per user per year

06 Interest accrual revenue

Cardo will be using a white label model for the first few years. Cardo will deposit cash into a traditional bank expecting 5% interest rate environment with a 30% commission going back to the bank.

As the business becomes more established, we are committed to exploring more revenue streams and refining existing models to maximise impact and profitability. This is explored in the expansion forecast section.

6.3.4 Surface level system map

The system map below shows where each revenue stream goes, ie ,service and what it contributes to.

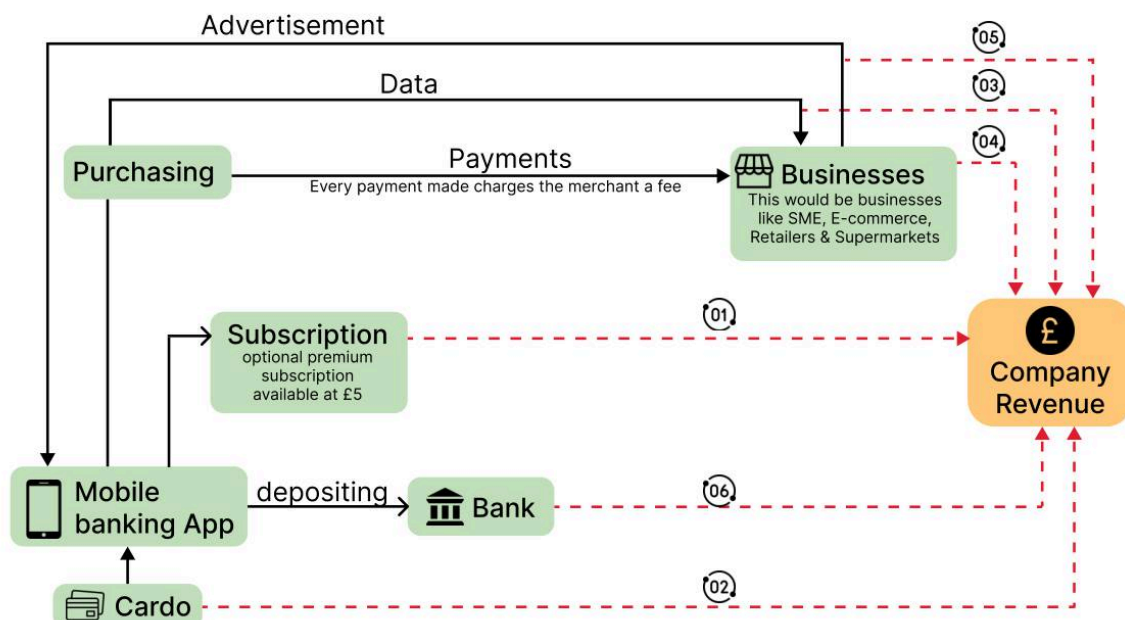


Figure 6.3.4.1 - surface level system map to represent revenue streams

6.3.5 Major revenue stream - interest accrual

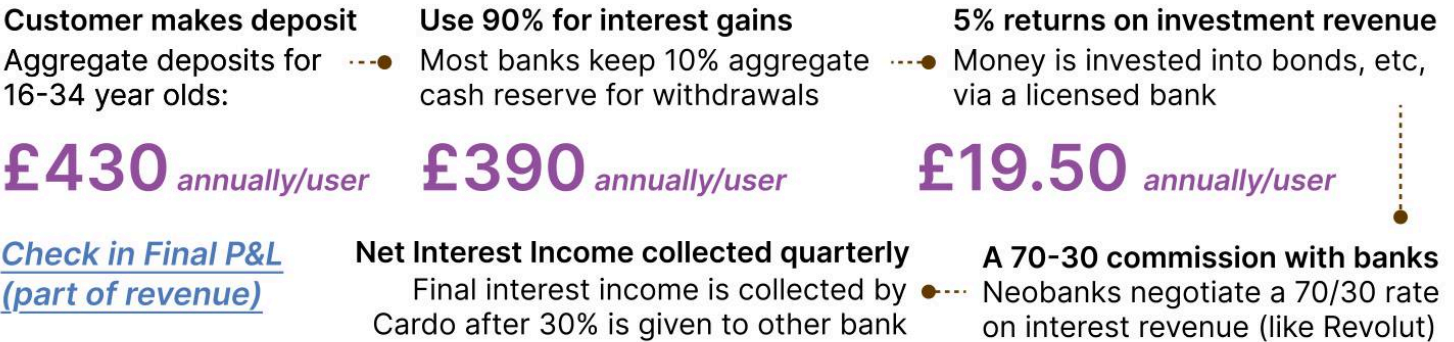


Figure 6.3.5.1: Interest Accrual references in market opportunity

6.4 Cost structure

Cardo will start off as a white label neo bank , which means it will use the infrastructure of an existing traditional bank i.e Lloyds bank and bank, this is due to the attainment of a banking license (this will be further explained in future expansion section). Below details the Most important costs associated with our business model with a focus on the 1st 6 months to a year of Cardo.

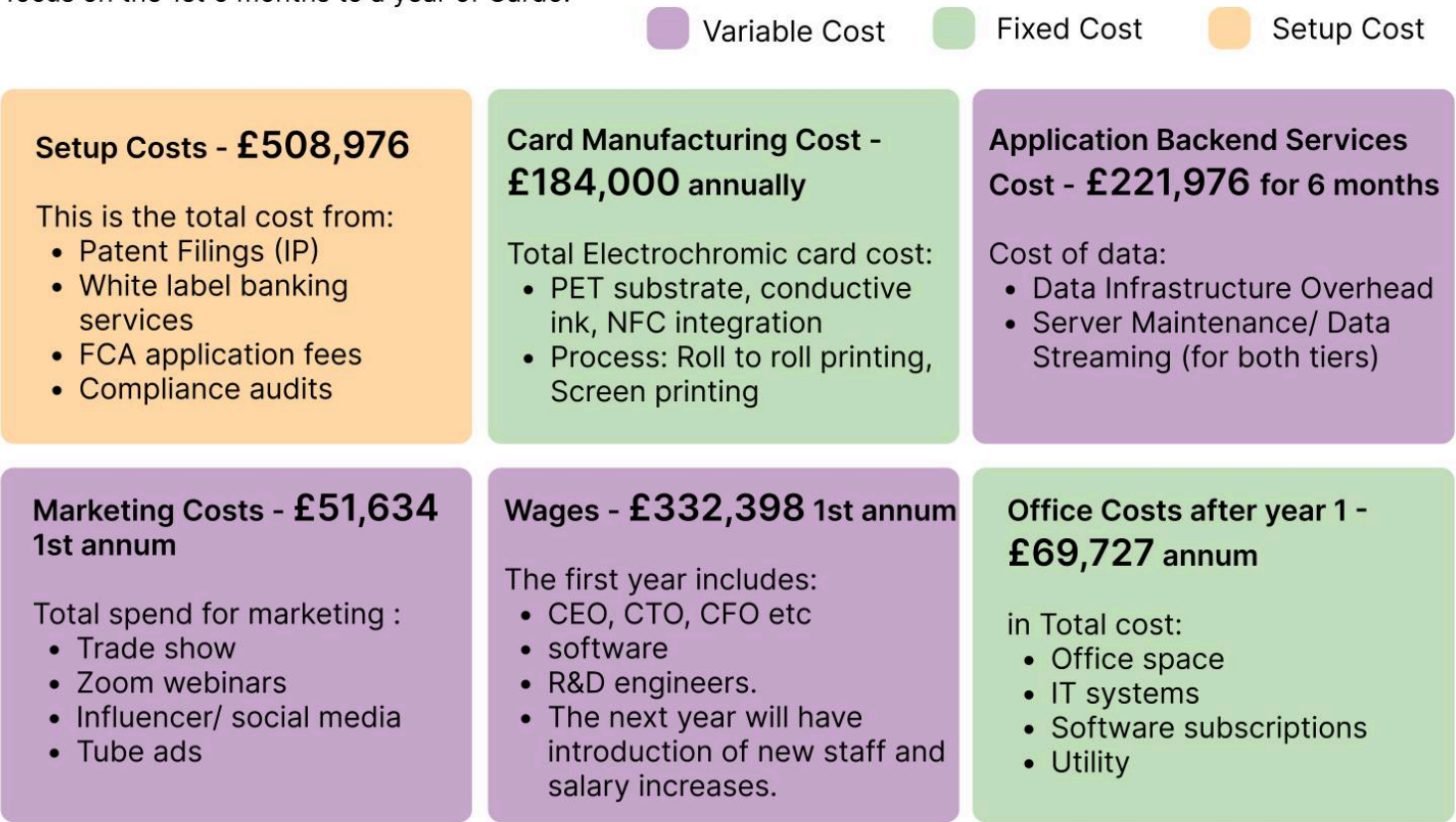


Figure 6.4.1 Cost Structure

Further Cost breakdown of the the cost structures can be found in the finances document under 'spending & expenses', 'Sales & Marketing' and 'COGS'.

6.4.1 Cost to User Comparison

Monzo is a leading neobank with the majority of users between ages of 25-35 (similar to our target group of young professionals).[1] Below is a comparison of Cardos key customer revenue schemes compared with the cost price for monzospremium features.

Cardo

Table 6.4.1.1: Cardo Cost to user

	Cost	Key word
Premium	£5	
Electrochromic card	£10	"Functional"

Monzo

Table 6.4.1.2: Monzo Cost to user [2]

	Cost	Key word
Premium	£15	
Metal Card	£50	"Aesthetics"

6.5 Expansion Forecast

After our business model, it is essential to conduct a forecast on the expansion of Cardo. This will be divided into 3 stages over the next 5 years, considering growth into other segments and regions:

- 1. **Initial** roll-out in London (year 1)
- 2. **Diversifying** to other customer segments and metropolitan areas in UK (year 1-3)
- 3. **Expanding** to other EU countries (year 3-5).

Initial roll-out (year 1)	Diversification (year 1-3)	EU Expansion (year 3-5)
Pilot program launched in London. Marketed to young adults, local affiliate partnerships.	Growing user base to other cities: Manchester, Edinburgh, diversifying revenue streams	Growing user base to other cities: Manchester, Edinburgh, diversifying revenue streams



Figure 6.5.1: Map of London



Figure 6.5.2: Map of England



Figure 6.5.3: Map of Europe

SMEs



Figure 6.5.4: SME

Elderly



Figure 6.5.5: Elderly stock photo

6.5.1 Initial roll-out - alpha launch in London (year 1)

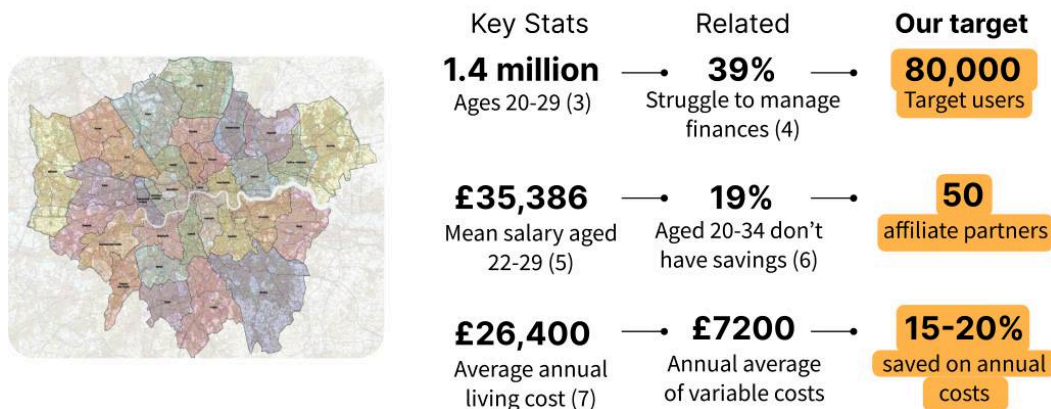


Figure 6.5.1.1: Year 1 Targets

Understanding some key figures to analyse, it is essential to 3 key metrics:

- 1. **The number of young adults to target** by understanding who struggle with finances. Take into consideration other roll-out programs as well (Monzo case study)
- 2. **The mean salary** for those aged 22-29, to understand how a variety of affiliate partners can be of value of both to the consumer, and to business alike.
- 3. **The average annual living cost**, to understand how the budgeting insights can be of help

6.5.2 Monzo Year 1 Growth (Case Study)



Figure 6.5.2.1: Monzo Beta

Monzo Beta launched

In 2015, Monzo Beta is launched as a pilot program in UK. Users are given pre-paid debit cards via Wirecard to save on costs. [9]

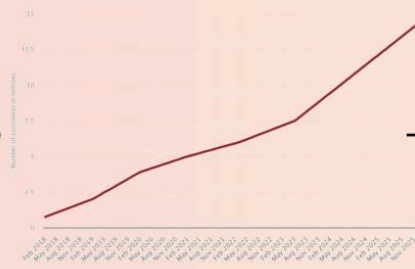


Figure 6.5.2.2: Monzo userbase growth (2015 - 2024)

Monzo user growth 2015-2024

Growth is recorded at approximately 260,000 new users at the end of year 1 in 2016. [9]

“App Bank Mondo Crowdfunds £1M In 96 Seconds” - Forbes

Monzo launches crowdfunding campaign 2016

Once userbase is grown, Monzo (known as Mondo at the time) launches a crowdfunding campaign [9]

Assumption Validation

80,000 target users

As seen from prior case studies of Monzo, and analysing the opportunity in the market, this validated our assumption that the target users at the end of year 1 will be 80,000.

50 affiliate partners

From our market opportunity and sales channel, we realised there was plenty of scope of approaching large scale businesses, and other SMEs for creating affiliate partners.

15-20% on annual costs

The 50-30-20 rule is a staple in budgeting practises. The smart budgeting insights and targeted ads should reinforce this model. Through our research, we found that the people spend >30% on wants.

Cost Breakdown

After validating our assumptions, the following graph shows a cost breakdown. Key Factors:

- **68% in data infra.** - Most early costs dedicated towards perfecting ML model and real-time data analysis
- **12% in card** - Since much of the manufacturing overhead is incurred in set up, costs are saved in year 1.

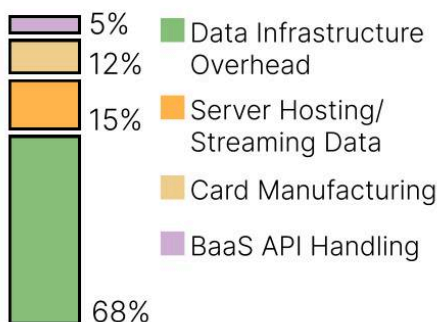


Figure 6.5.2.3 Cost breakdown of 1st year

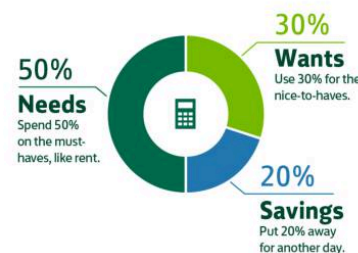


Figure 6.5.2.4: Lloyds breakdown of budgeting [10]

6.5.3 Diversification (year 1-3)



Key Stats	Related	Our target
5.3 million Ages 20-29 in UK	39% Struggle to manage finances	200,000 Target users
5.6 million SMEs in the UK [11]	350,000 companies in affiliate programs [12]	2000 affiliate partners
3.5 million Ages 36-60	3% Of all adults to switch banks	50,000 middle aged people

Figure 6.5.2.3 Cost breakdown of 1st year

Understanding some key figures to analyse, it is essential to 3 key metrics:

1. **The number of young adults** to target in the UK is **5.3 million**, out of which a realistic number to take is 200,000 target users
2. The number of affiliate partners is **2000**, which SMEs and other companies are a part of.
3. Diversifying our revenue stream to include middle aged people is essential to ensure business sustainability. Out of **3.5 million**, **3%** are likely to switch, hence leading to a target of **50,000**

6.5.4 Year 3 - Acquisition of Banking License

After operating as a white-label bank for the first 2 years (working via an existing banking system), we plan to acquire a full banking license to operate compete with existing banks.

TOTAL Cost: £7,600,000

Increased Competition

This is because we assume other banks will provide similar budgeting features we are during this period to compete, requiring us to certify the Cardo brand and attract more capital to manage competition.

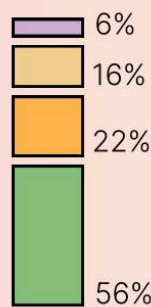


Figure 6.5.4.1: Cost breakdown of a banking license

Breakdown

- £4.3 million is the initial capital required by the PRA to show financial stability. [15]
- According to Deloitte, approx. £1.5-2 million is necessary for consulting on legal compliance [14]
- Expert Alastair Holt places core banking system to cost £1.2 million, on the lower end [13]

Revenue Diversification

5000 affiliate partners

Through our research, we found that SMEs benefit greatly from large affiliate programs. Ones we will target to provide ads include 3 key ones: **Amazon, Shopify, & CJ Affiliate.** These cover the following [16]:

- SMEs - Local ads for users
- Supermarket Chains - Bigger deals to cut costs (alternate revenue)
- E-commerce chains - Deliveroo, UberEats, Asos, etc.


66,000 SMEs



Superchains: Tesco's, Dell, Sainsbury's

50,000 middle- aged people

According to research conducted from market opportunity, it was found that 3% of adults are likely to switch banks. Using this statistic, we set a target for targeting 50,000 users from ages 36-60 by the end of year 3.

This is validated with customer segments of other banks: Revolut - 73% are 16-34, rest are 34+.

6.5.5 EU Expansion (year 3-5) - scaling after 3 years

Meeting EU Financial Regulations for expansion

To expand into the EU, there are key regulations that must be met. These include the following:

- **European Economic Area (EEA) Authorized Representative Model** - for limited neo bank permissions
- **E-Money Institution License** - license under PSD2 for conducting transactions via accounts
- **Basel III-aligned requirements** - meeting minimum capital requirements [17]



Licensing Electrochromic Card technology to other banks

Leasing technology to other banks and companies may be useful. This was validated by Louis, the product owner from Revolut, who mentioned how other neo banks may be interested in the product if it gains traction.

He also stated the importance of selecting viable 'price models' for licensing the technology. Future R&D must be conducted to address the importance of a tiered service to other banks perhaps.



Louis Cutner, 24

"Think of partnerships with banks. They may be interested"

7 Assumptions

7.1 Idea Hypothesis Canvas

The key assumptions identified throughout this project have been collated here and the Idea Hypothesis Canvas has been used to systematically evaluate and subsequently validate them.

Problem	Solution	Beachhead Customer	Business Model
Managing personal finances is overwhelming for many, and current budgeting tools are often complex and disconnected from real-time spending. Current Solution to Problem: Are spreadsheets and static budget apps in which you have to manually update them.	A budgeting card with electrochromic technology that provides real-time spending feedback paired with an app offering custom budgets and detailed spending insights. Unique Selling Point: An intuitive visually engaging physical card and an app that deeply understands user spending.	Young professionals that are aiming to gain financial literacy and control and individuals who struggle to budget effectively due to lack of time or accessible tools. Size of Potential Market: From our bottom up approach the approximate size is 340,000 people in the first 3 years.	Consumers: - Subscription fess - Card Sale Revenue Business - Affiliate marketing - Data Sales - Interest Accrual - Interchange Fees Price for Customers: For the premium model of the app it will cost £5 per month. For businesses data per user per year is £31

Untested Assumptions

- Consumers find existing tools like apps and spreadsheets too disconnected from their real-time spending habits to be useful. - Users want budgeting tools that are intuitive and require minimal effort to understand and implement. - A significant portion of consumers struggle with managing their finances due to poor financial education, making them more receptive to innovative solutions.	- The card's electrochromic technology can be produced affordably at scale. - The app will have the ability to categorise spending intelligently (e.g., distinguishing one-off payments vs recurring expenses). - The integration of a physical card with an app will appeal to both tech-savvy and less tech-savvy users.	- Young professionals will find visual spending feedback compelling enough to adopt the card. - Young adults are particularly vulnerable to poor budgeting habits, making them a prime audience for tools that simplify and automate budgeting.	- Businesses will see sufficient value in purchasing anonymised consumer spending data. - Consumers will be willing to pay for premium app features - Revenue from the business and consumer streams will sustain manufacturing and operational costs. - Payment networks (e.g., Mastercard) will support the integration of electrochromic technology without resistance.
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7.2 Categorising Assumptions

Internal	External
The electrochromic card can be manufactured affordably and reliably at scale. The business can comply with data protection regulations like GDP The technology behind the electrochromic card is scalable and durable enough for everyday use. The card and app can be distributed efficiently to target customers. Consumers will pay £5/month for premium app features. The app will be able to categorise spending intelligently. Our software engineers will create a machine learning algorithm to deliver tailored insights.	Financial stress and rising living costs will continue to push people towards budgeting tools Suppliers will provide electrochromic components at competitive prices Affiliate partners will be willing to collaborate and promote the card. Payment networks (e.g., Mastercard) will support the card. Businesses will see actionable consumer insights as worth paying for. Young professionals will find visual spending feedback compelling enough to adopt the card. The UK economy remains relatively stable

Assumptions were categorised into internal and external factors to better understand their impact on the business. Internal factors focus on aspects that the business can control, whilst external factors include elements outside the business's control.

The assumption map uses the ICE framework to prioritise key business assumptions by scoring Impact, Confidence, and Ease (1-10). These scores were then multiplied together to determine priority:

- **Impact:** Rated based on importance to business outcomes.
- **Confidence:** Reflects how certain we are about the assumption
- **Ease:** This is based on resources and complexity required for validation

Our group assigned these ratings collaboratively, using discussions and expert input to evaluate each assumption. Impact was determined by assessing how critical the assumption is to the project's success, while Confidence scores reflected the level of evidence supporting each assumption. Ease scores were based on the perceived difficulty of validating the assumption, factoring in time, resources, and expertise. This structured scoring ensured a balanced and informed approach to prioritising assumptions for immediate validation.

7.3 Assumption Mapping

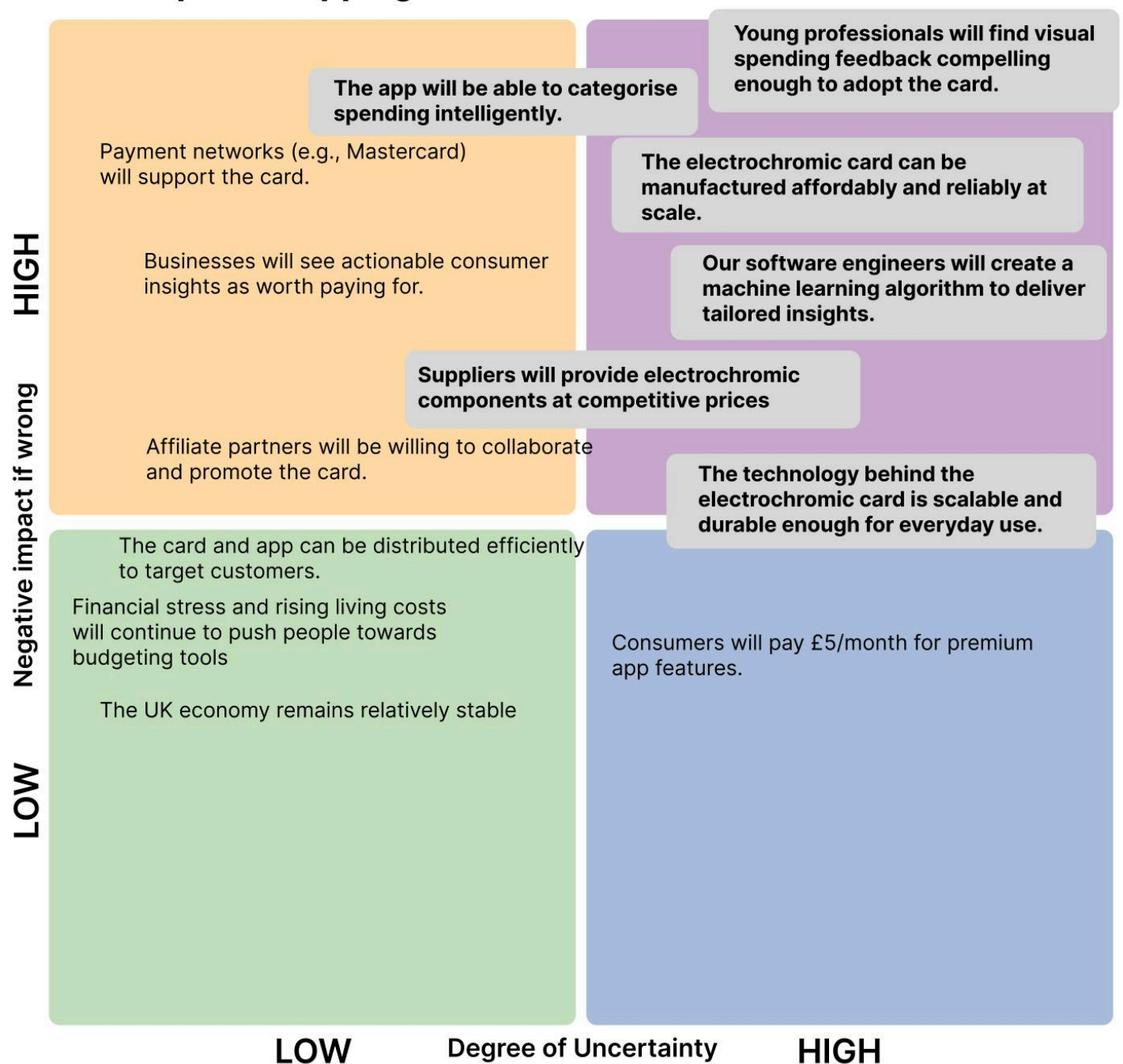


Figure 7.3.1 - Assumption Mapping Graph

7.3.2 3 Key Assumptions

1 Young professionals will find visual spending feedback compelling enough to adopt the card.

2 The electrochromic card can be manufactured affordably and reliably at scale.

3 Our software engineers will create an algorithm to categorise spending and deliver tailored insights.

From our assumption mapping, several key assumptions were identified. While more than three were highlighted, similar assumptions were combined for clarity. For instance, multiple assumptions regarding the electrochromic card were combined into a single assumption due to their overlap. Similarly, assumptions about the app that shared similarities were combined.

7.4 Assumption Validation

To validate our key assumptions, extensive research and communication with industry experts were carried out, ensuring a robust justification for each assumption.

1 Young professionals will find visual spending feedback compelling enough to adopt the card.

Louis Cutner



Graduate Product Owner at Revolut

"When I talk to my friends about using contactless payment options, including Apple Wallet, they say they're open to trying something new but feel that most current cards lack **convenience** and **appeal**. I think a card with new technology and a sleek, aesthetic design wouldn't just add functionality but would also create a real 'wow' factor. For example, at Revolut, I got a metal card with a sleek silver finish, and it felt so luxurious almost like silk. My friends were always asking how they could get one because they were so impressed by the design. **I really believe that combining functionality with standout aesthetics makes a product incredibly desirable.**"

2 The electrochromic card can be manufactured affordably and reliably at scale.

Research into View Inc, a company specialising in electrochromic windows, demonstrates the feasibility of manufacturing an electrochromic card. Electrochromic materials are competitively priced, scalable production processes are well established, and the widespread adoption of this technology across industries suggests strong market acceptance for consumer products.

view

In 2022, View Inc. reported revenues of **\$101 million**, a **37%** increase from the previous year, demonstrating growing market acceptance. [1]

The company has a production capacity of about **1 million** square feet of dynamic glass per year, showcasing its ability to scale manufacturing processes.[2]

Electrochromic glass typically costs between **\$50 and \$100** per square foot.[3]

3 Our software engineers will create an algorithm to categorise spending and deliver insights.

- ARIMA Models: Used for time series forecasting, ARIMA predicts spending patterns, aiding in budgeting and financial planning.[4]
- AWS Kinesis & Apache Kafka: Real-time data platforms that process transaction data, enabling instant categorisation and analysis.[5]
- NLP Techniques: Analyse transaction descriptions to accurately classify spending categories.[6]
- Supervised ML Algorithms: Models like Decision Trees and Neural Networks predict spending categories using labeled data, improving accuracy over time.[7]
- Unsupervised Clustering: Methods like K-Means group transactions, uncovering patterns and emerging categories.[8]

The research highlights various existing algorithms, demonstrating that our software engineers can develop a machine learning algorithm capable of accurately categorising spending and delivering personalised insights.

8 Sales Channels

8.1 Bulls-Eye Framework

Our solution targets young professionals, but is a budgeting tool for all ages and so it is imperative we cater our sales strategy and marketing scheme to appropriately. To initially brainstorm existing sales channels we used a 'Bulls-eye Framework'.

From an initial 20 we narrowed down to 7 ideal sales channels, from this we used a RICE framework to consider reach, impact, confidence and effort required for each. As a result we selected a top 4 which we will analyse which are not only effective but scaleable. (please see appendix for RICE framework breakdown).

Despite Trade Shows and Webinars performing slightly worse for than Email Marketing, Trade Shows provides a highly profitable method of exercising our B2B service of selling consumer data as shown in its respective sales funnel. Additionally, webinars help us to directly interact with users and foster a community. (Please see Appendix for breakdown).

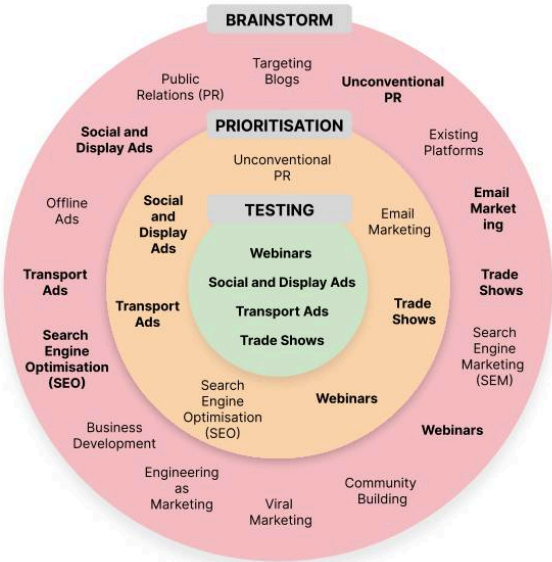


Figure 8.1: Bulls-eye Framework

8.2 Sales Channels

Our chosen sales channels highlights the most successful methods to initially roll out our product. The combination of Webinars, Social Media Ads, Transport Ads, and Trade Shows provides the perfect balance between reach, engagement, and scalability. These channels were selected for their ability to effectively reach individual consumers, ensuring broad visibility and strong engagement.

Customers

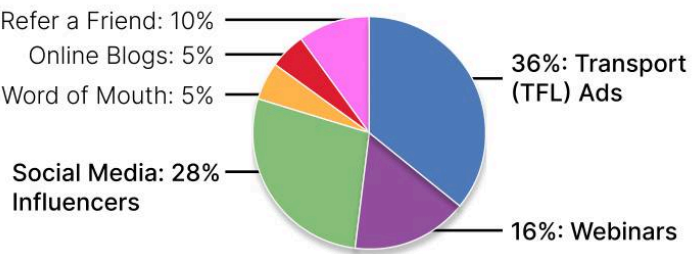


Figure 8.2.2: Sales Channel Breakdown - Customers

Businesses

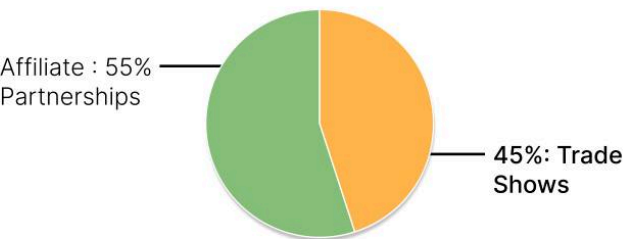


Figure 8.2.2: Sales Channel Breakdown - Businesses

Our affiliate partnerships leverage relationships with large-scale businesses, which have extensive networks of Small and Medium Enterprises (SMEs) [1]. These partnerships act as trusted intermediaries, driving awareness of our product among SMEs who are typically harder to reach directly. By aligning with these larger players, we unlock access to a secondary audience, broadening our customer base. In addition to Trade Shows this will account for an additional 55% of Business Acquisition.

Key Assumption

From our premium subscription, we assume a churn rate starting at 30% in month 1, gradually decreasing to 2% by month 12, following an exponential decline. This reflects our strategy to enhance user retention through improved customer experience, targeted engagement campaigns, and the added value of premium features, fostering long-term loyalty.

8.2.1 Zoom Webinars

In Year 1, we plan to host **five targeted webinars** tailored to young professionals, highlighting the capabilities of our budgeting tool and insights it provides. With webinars proven to generate 2–3 times more leads than other content formats [2], this channel will serve as a cornerstone for engaging our audience. Each session will feature expert guest speakers, interactive Q&A opportunities, and exclusive offers, designed to build a community of trust and awareness while staying cost-effective.

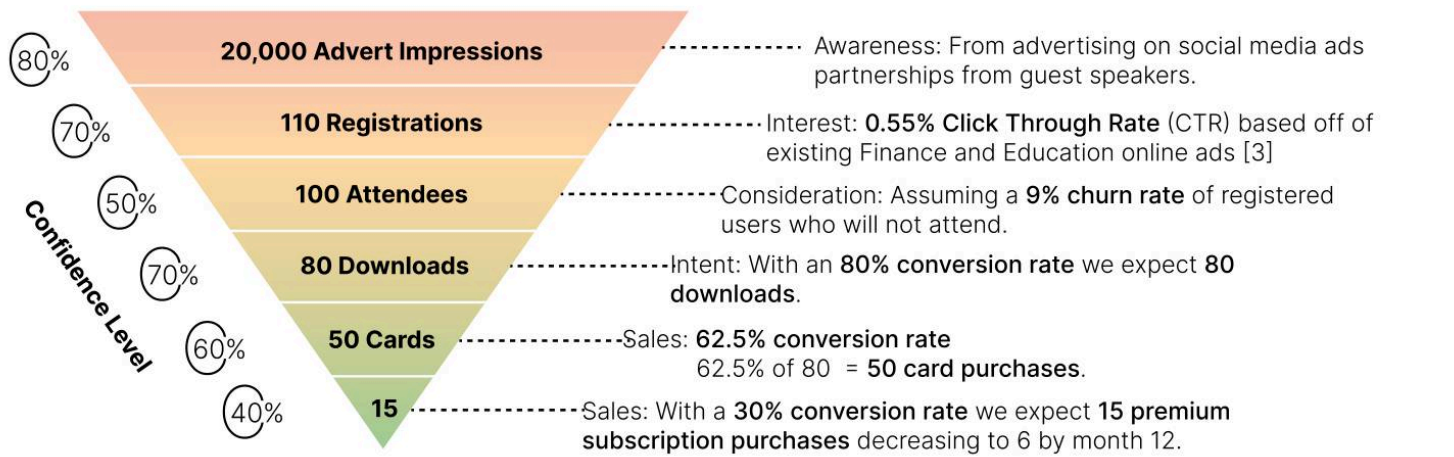


Figure 8.2.1.1: Sales Funnel - Zoom Webinars

In Year 2, the webinar budget will increase from £1,427 per event in Year 1 to £2,500, enabling enhanced production quality with multi-day events, upgraded Zoom branding, and prominent guest speakers. Collaborations with industry experts will deepen, increasing attendance to 500 and boosting card purchases and premium subscriptions by 10%, reducing CAC from £22.54 to £8.93. In the latter half of Year 2 and into Year 3, we plan to introduce in-person conferences with a £5,000 budget per event, fostering stronger community engagement and conversion rates.

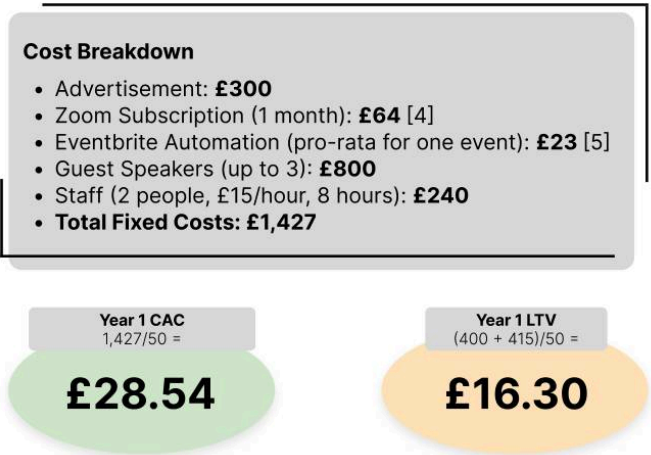


Figure 8.2.1.2: Cost Breakdown - Zoom Webinars

These conferences will focus interactive Q&A sessions debugging budgeting myths to educate potential users along with live demonstration which we anticipate will significantly boost engagement and conversion rates. The hands-on nature of these events will offer customers a more immersive experience for users to see how Cardo works aiming to increase card purchase rates to up to 80%. This phased strategy will strengthen customer acquisition and loyalty as our product evolves.

Users will be given code **WEB20** for **20% off their card purchase** along with being incentivised them and others to join Cardo with a **refer a friend scheme** to receive your first month of premium free.

8.2.2 Social Media and Influencer Marketing

We are partnering with TikTok influencers, both micro and macro, to connect with audiences interested in budgeting and financial literacy. As part of this, we will provide influencers with our card along with one month free premium access, allowing them to share their authentic experiences with their followers. By leveraging their engaged fanbases, we aim to further strengthen our brand's following and awareness on how easy money management can be with our product. This approach ensures ads effectively tailored to target users.

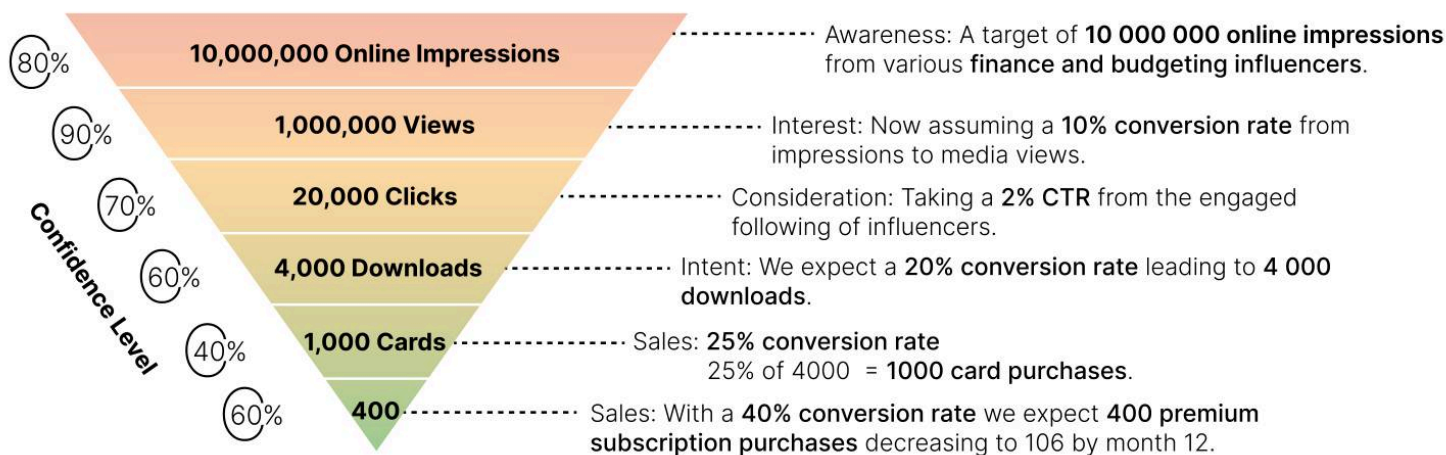


Figure 8.2.3: Sales Funnel - Social Media and Influencer Partnerships

In Years 2 and 3, we aim to increase this budget to £10,000 and £20,000 respectively. As we collaborate with a higher proportion of macro influencers, we will expand our reach and increase brand visibility. Utilising their larger followings will amplify our message and showcase real user success stories on a larger platform, ensuring sustained growth and market penetration.

Table 6.2.2: Influencer Price Breakdown

	No. of Followers	Price Per Post
Mega Influencers	1 000 000 +	£10 000 +
Macro Influencers	100 000 - 1 000 000	£3 000 - £10 000
Mid Influencers	50 000 - 100 000	£1 500
Micro Influencers	10 000 - 50 000	£500

Cost Breakdown

- 4 Micro (£2,000) + 2 Mid (£3,000) : **£5,000** [6]
- **Total Fixed Costs: £5,000**

Year 1 CAC

5,000/1,000 =

£5.00

Year 1 LTV

(8,000 + 9,150)/1,000 =

£17.15

Figure 8.2.3: Cost Breakdown - Social Media and Influencer Marketing

8.2.3 Tube Ads

Our tube advertising strategy targets high-footfall stations such as King's Cross St Pancras, Euston, and London Bridge, which are frequented by professionals and students, aligning with our core audience. Starting with affordable 6-sheet ads [7] run in 2 week intervals costing £2,000 per month, we aim to reach 5,000,000 daily passengers across the busiest lines. By focusing on ABC1 commuters aged 15–34 we project 640,000 ad views, leading to 12,800 searches and 2,500 app downloads. This targeted campaign allows us to build notoriety young professionals whilst minimising cost.

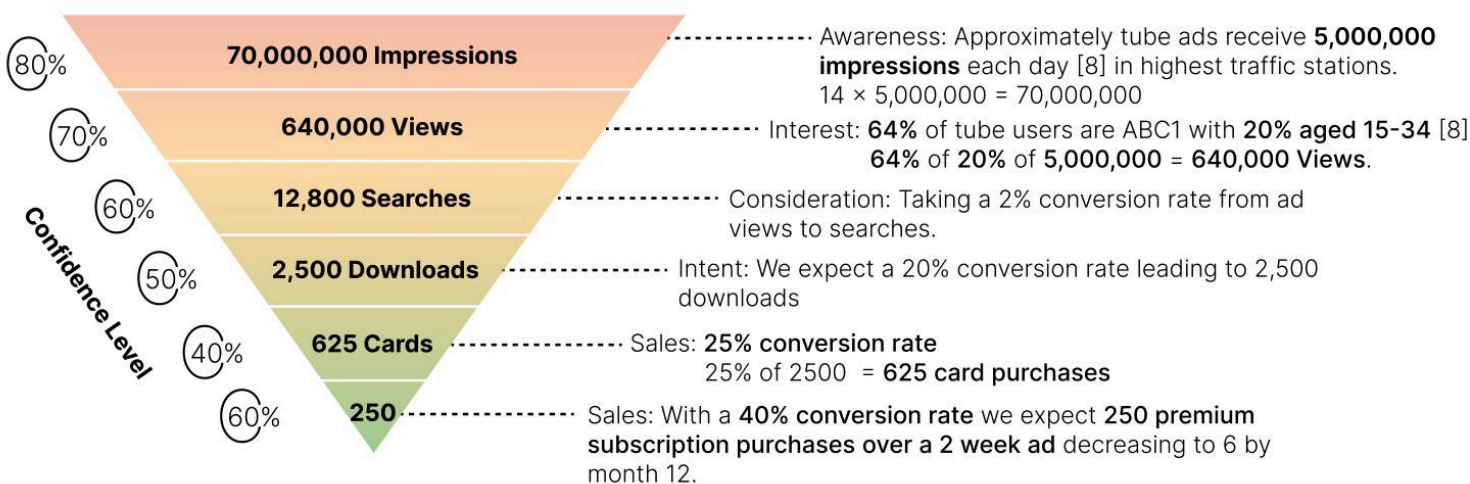


Figure 8.2.3.1: Sales Funnel - Tube Ads

On tube adverts code **TUBE20** to receive 20% off users card with this ad changing to **TUBE10** for 10% off from Year 2.

We plan to scale up our tube advertising with a £3,000 monthly budget, enabling us to include prominent digital screens and 48-sheet ads for broader visibility in key stations. Interactive features, such as QR codes, will further encourage immediate app downloads and card purchases. The larger, eye-catching formats will not only enhance brand visibility but also create a more memorable impression among commuters.



Figure 8.2.3.2: Cost Breakdown - Tube Ads

8.2.4 Trade Shows

We aim to showcase our platform at Big Data LDN [9] in month 6, as a prime event for data-driven businesses to be showcased. With 77% of executive decision-makers finding at least one new supplier at trade shows [10], this sales channel provides a golden opportunity to present our value proposition to our target audience. By this point, we project reaching 20,000 users, enabling us to market data reports at £4,000 each (20p per user) [11]. These reports will offer detailed insights to companies regarding financial spending habits of users, demonstrating the value of our unique dataset.

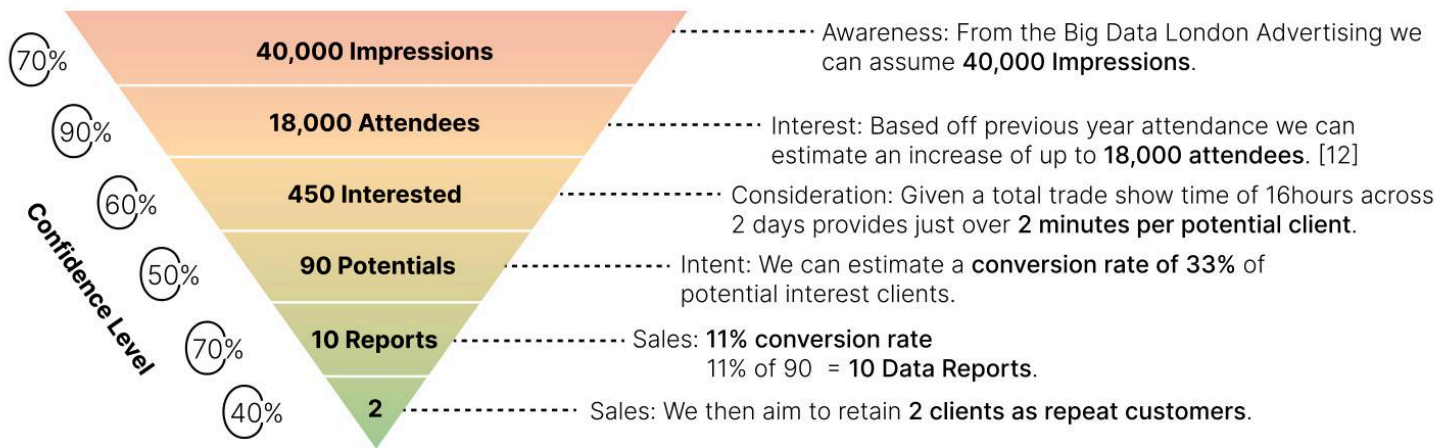


Figure 8.2.4.1: Sales Funnel - Trade Shows

Building on the success of our first trade show, we plan to attend three trade shows in Year 2 and another three in Year 3, scaling our reach and visibility. For the first year, at a cost of £300/m², [12] the cost for a 3m by 2m booth would be £1,800. In Year 2, we plan to increase our budget to £3,800 to expand booth size and improve materials to enhance our professional presence. By Year 3, this will rise further to £5,000, enabling us to create an interactive and engaging booth experience. This will not only house more staff but help us reach our goal of a 50% sales conversion rate.



Figure 8.2.4.2: Cost Breakdown - Trade Shows

As our user base grows, we aim to increase the value of our data reports to 25p per user in Year 2 and 30p per user in Year 3, with datasets expanding to hundreds of thousands of users. These strategic enhancements will position us as a leader in delivering high-value data insights, fostering stronger relationships with key industry players, and maximizing the revenue potential of each trade show.

9 Profit & Loss

[See full P&L spreadsheet here](#)

9.1 Operational Costs

9.1.1 Electrochromic Card Costs

The production of the electrochromic card will be outsourced, with materials sourced separately from a dedicated supplier to reduce costs. The current cost estimates assume that conductive electrochromic ink may require two layers instead of one due to ongoing R&D. This prediction is subject to change depending on further research outcomes, which could lead to cost adjustments.

Table 9.1.1.1: Electrochromic Card Manufacturing (Cost/per card)

Electrochromic Card Manufacturing (Cost/per card)				
Material	Setup	Year 1	Year 2	Year 3
PET Substrate	£0.32	£0.24	£0.22	£0.22
PSS Conductive Electrochromic Ink	£0.40	£0.68	£0.60	£0.56
Silver Nanowire Layer	£0.63	£0.48	£0.40	£0.38
Polymer Gel Electrolyte	£0.24	£0.20	£0.18	£0.18
Adhesive Polymer Spacer	£0.12	£0.10	£0.10	£0.10
NFC Integration	£0.65	£0.55	£0.52	£0.52
Flexible Microcontroller Unit	£0.79	£0.70	£0.65	£0.65
Manufacturing	-	-	-	-
Roll-to-Roll Printing	£0.47	£0.43	£0.42	£0.42
Screen Printing	£0.32	£0.30	£0.30	£0.28
Total:	£3.94	£3.68	£3.39	£3.31

The cost per card decreases steadily over three years as production processes improve and economies of scale are achieved, starting at **£3.68 in Year 1**, reducing to **£3.39 in Year 2**, and **£3.31 in Year 3**. These reductions are driven by optimised material costs, including components like conductive electrochromic ink and microcontrollers, alongside improved manufacturing efficiencies such as roll-to-roll and screen printing.

9.1.2 External Card Costs

In addition to production costs, shipping, assembly, and export duties reflect the scaling of operations. Shipping costs rise from **£8,150 in Year 1 to £63,500 in Year 3** as production volume grows. Assembly costs also increase, starting at **£13,688** and reaching **£92,000** by **Year 3**, reflecting labour and integration processes. Export duties remain relatively low, increasing from **£3,000 to £18,000** over the same period, due to outsourcing to countries with lower tariffs such as **Vietnam, India, and China**. The full breakdown of these can be seen in the “Employees and Expenditure” workbook of our full P&L spreadsheet.

9.1.3 App Costs

Table 9.1.3.1: App Services Costs

The app costs cover backend services critical for Cardo's operations, scaling with user growth and data usage:

- **Data Streaming & Analysis:** Enables real-time spending insights using AWS Kinesis and ARIMA.
- **Storage & Hosting:** Ensures secure, scalable infrastructure with AWS Cloud Database services.
- **Security & Encryption:** Manages sensitive data securely through AWS Key Management.

App Services Backend		
AWS Kinesis & ARIMA	Per Stream/Per Hour	£0.038
	Data Ingested (GB)/ User	£0.075
	Data Retrieval (GB)/ User	£0.048
AWS Cloud Database Services	Data Stored/ (GB)/ User	£0.116
	Server Hosting per Month/ User	£0.55
	Server Hosting per Month/ User (Premium)	£1.39
AWS Key Management	Per Key/ Per Month	£1.00

The app costs also include Banking-as-a-Service (BaaS), a critical component of Cardo's backend infrastructure. As Cardo operates as a white-label neo bank without its own banking licence, BaaS providers enable access to essential banking functionalities such as account management, payment processing, and regulatory compliance. This allows Cardo to offer core financial services seamlessly while focusing on user experience and innovation.

9.1.4 Employees

In Year 1, total employee salaries amount to £279,450, increasing to £608,139 by Year 3 as more employees are onboarded and salaries grow by an assumed 5.4% annual increase. Key roles include leadership positions such as CEO, COO, and CTO, each starting at £35,000 during the setup period, rising to £40,982 by Year 3. Additionally, technical positions like Software Engineers and R&D Engineers are brought in, aligning with scaling production and development. For further details on variables such as pensions and national insurance check the the full P&L spreadsheet.

Table 9.1.4.1: Annual Salaried Worker (Employee) Expenditure Breakdown

Salaried Worker (Employees):				
Employee	Setup	Year 1	Year 2	Year 3
CEO	£35,000	£36,890	£38,882	£40,982
COO	£35,000	£36,890	£38,882	£40,982
CTO	£35,000	£36,890	£38,882	£40,982
CFO	£35,000	£36,890	£38,882	£40,982
CMO	£35,000	£36,890	£38,882	£40,982
Software Engineer	-	£32,500*2	£34,225*3	£36,105*4
R&D Engineer	-	£30,000	£30,000*2	£31,620*3
Financial Advisors	-	-	£30,000*2	£31,620*2
Customer Service	-	-	£24,386*2	£25,703*4
Total / Month:	£18,265.63	£27,699.80	£42,744.70	£59,504.72

9.1.5 Office Costs

The office costs cover essential infrastructure to support Cardo's growing team. Each desk setup (£6,000/year) includes chairs, computers, monitors, and peripherals, with desks increasing from 8 in Year 1 to 18 in Year 3. IT system costs rise from £1,180/month to £2,530/month, ensuring robust operations. Software expenses grow from £180.60/month to £312.80/month, while utilities, including electricity and internet, increase from £450/month to £600/month. Overall monthly office costs rise from £5,810.60 in Year 1 to £12,442.80 in Year 3, supporting team efficiency. For a full breakdown, see the Employees & Expenditure workbook in the full P&L spreadsheet.

9.2 Setup Costs

Table 9.2.1.1: 6 months Setup Cost Expenditure Breakdown

Setup Costs	
Patents	£5500
White Label Banking	£150,000
FCA Application Fee	£15,000
Ongoing Compliance Audits	£30,000
Data Infrastructure Overhead	£174,600
Electrochromic Card	£39,400
BaaS API Handling	£46,500
Server Hosting/ Data Streaming	£33,288
Server Hosting/ Data Streaming (Premium)	£14,688.33
Total:	£508,976.33

The setup costs for Cardo include essential expenses required to launch operations and build a scalable infrastructure.

- **Patents** cover intellectual property protection, ensuring exclusivity over the electrochromic card technology.
- **White Label Banking** enables Cardo to operate as a neo-bank without a direct banking licence, providing core banking infrastructure through a third-party.
- **Ongoing Compliance Audits** ensure Cardo meets regulatory standards, particularly as we operate under a BaaS model.
- **Data Infrastructure Overhead** reflects backend services, such as AWS hosting, streaming, and security, which scale with user data and activity.
- The assumption of acquiring 10,000 users in our 6-month setup phase also includes providing electrochromic cards for free.
- BaaS API Handling covers the integration of banking APIs, while server hosting and data streaming provide the infrastructure for secure and seamless transactions.

9.3 Costs Of Goods Sold

The Cost of Goods Sold (COGS) includes all direct costs associated with delivering our products and services, ensuring smooth operations and functionality. Key components include **Data Infrastructure Overhead**, which covers expenses for maintaining servers, databases, and critical networking systems. **Electrochromic Card** represents costs tied to the production or acquisition of specialised hardware integral to the product. Additionally, **BaaS API Handling** includes fees for integrating Banking-as-a-Service APIs, essential for backend operations and financial services. Lastly, **Server Maintenance / Data Streaming** costs are divided into standard and premium tiers, reflecting expenses for server upkeep and data streaming services, with the premium tier offering enhanced features. For Year 1, the total COGS amounted to £880,900.84, and for Year 2, it increased to £1,655,106.17, reflecting growth in operational scale and product demand.

9.4 Revenue

The revenue breakdown highlights Cardo's key income streams:

- **Interest Accrual:** Earnings from interest on user deposits and balances.
- **Data Insights:** Revenue from selling anonymised user spending insights to third parties.
- **Interchange Fees:** Fees earned per transaction when users make payments with the Cardo card.
- **Sales & Marketing:** Income from partnerships and promotional collaborations.
- **Advertisement:** Revenue from targeted ads leveraging Cardo's user base.

Table 9.4.1:Revenue Breakdown

Revenue Breakdown				
Stream	Setup	Year 1	Year 2	Year 3
Interest accrual	68125	912909	5672918	15986219
Data Insights	77500	536873	1060813	1943092
Interchange Fees	2725	13214	26109	47825
Sales & Marketing	56554	283335	944965	3624570
Advertisement	48600	336671	665232	1218507
Total	253504	2082996	8370039	22820214

9.5 Marketing Costs

Our marketing strategy combines digital campaigns, influencer partnerships, and targeted events to drive engagement and growth. Social Media Influencers play a key role, with a varied mix of micro, mid, macro, and mega influencers each year to expand reach effectively.

Our first Trade Show is introduced to target business to sell consumer data in Month 10 after reaching 20,000 users. Until then, more cost-effective channels like Tube Ads and Webinars are prioritised. By Years 2 and 3, we scale investments in Tube Ads and Trade Shows, alongside an increased focus on influencer outreach to maximise acquisition.

Table 9.5.4.1: Annual Marketing Expenditure Breakdown

	Year 1	Year 2	Year 3
Zoom Webinars	£9,054	£14,800	£21,600
Tube Ads	£12,000 £2,000*6	£30,000 £5,000*6	£48,000 £8,000*6
Social Media Influencers	£28,000 14+14+0+0	£59,500 8+12+5+0	£101,500 26+14+3+3
Trade Shows	£2,580	£9,600	£10,800
Annual Total	£51,634	£113,900	£181,900

9.5.1 Monthly Marketing Expenditure

Below highlights that the first 6 months post-setup focus on rolling out initial advertising campaigns. Key **spikes** occur in **Months 14** and **16** in Year 2 with major social media campaigns in Months 20 and 24. with **significant peaks** in **Months 26, 28, 32, and 34** through Year 3 reflecting increased spending on targeted campaigns.

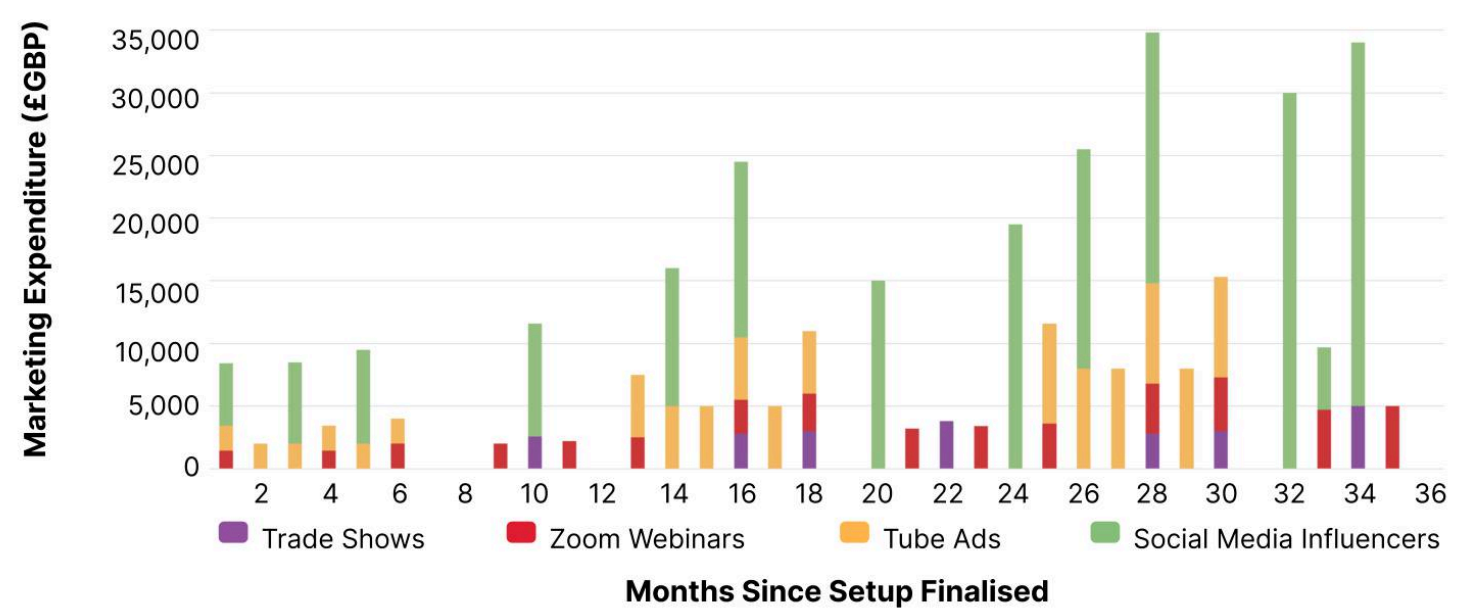


Figure 9.5.5.2: Monthly Marketing Expenditure Forecast

9.5.2 Customer Acquisition Cost (CAC)

Each sales channel has a distinct CAC and weighting averaged over each year dependent on overall contribution to customer acquisition.

Tube Ads maintain the lowest CAC at £3.20, with Webinar CAC decreasing over the years. Despite fluctuation in Social Media CAC, this is driven by a strategic mix of micro, mid, and macro influencers, ensuring a balance of cost, audience reach, and engagement. Organic channels are forecast to grow significantly, reaching 46% of our acquisition share by Year 3, enabling us to achieve a competitive weighted CAC of £3.64.

Table 9.5.4.1: BAC Annual Weighting

	Year 1		Year 2		Year 3	
	Weight	CAC	Weight	CAC	Weight	CAC
Zoom Webinars	19.8%	£25.24	17.4%	£22.85	10.4%	£22.65
Tube Ads	32.9%	£3.20	36.6%	£3.20	37.7%	£3.20
Social Media	27.3%	£4.25	26.0%	£5.77	31.8%	£5.73
Organic Sales	20.0%	£0.00	30.0%	£0.00	46.0%	£0.00
Weighted Sum	£7.21		£5.81		£3.64	

9.5.3 Customer Lifetime Value (CLTV)

Each user generates £10 in revenue per card, with a production cost of £3.68 in Year 1. By taking the churn rate average over one year, we assume 10% of users leave the premium subscription service annually, resulting in an average of 4 months on free membership and 32 months on premium membership over a 3-year (36-month) period. **Data hosting costs** amount to **£0.55 per month for standard users** and **£1.33 per month for premium users**, with premium user percentages weighted according to the sales channel. Finally, we include the profit from selling consumer data at **£0.20 per user per month**, contributing to the overall lifetime value calculation.

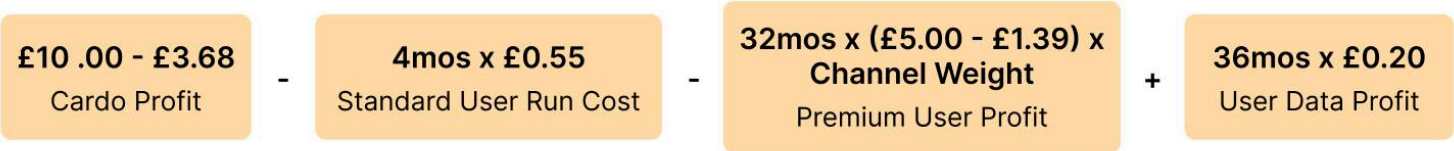


Figure 9.5.5.1: BLTV Calculation Breakdown

Each year, card manufacturing costs decrease, boosting CLTV. However, the growing share of users from organic sales, with a lower **4% premium conversion rate**, causes CLTV to decline over time. Despite this, as our weighted CAC decreasing at faster rate than that of our CLTV our ratio of CAC to CLTV increases, predicted to reach **1:6 by Year 3**.



Figure 9.5.5.2: Annual CLTV and CAC:CLTV Breakdown

9.5.4 Business Acquisition Cost (BAC)

Our acquisition strategy balances Trade Shows and Affiliate Marketing. Trade Shows drive early growth, while affiliate partnerships scale to 65% by Year 3 at a £0 CAC as an organic channel. This aims to reduce overall BAC to £127.29 by Year 3.

Table 9.5.4.1: BAC Annual Weighting

	Year 1		Year 2		Year 3	
	Weight	BAC	Weight	BAC	Weight	BAC
Trade Shows	45.0%	£258.00	40.0%	£176.36	35.0%	£168.75
Affiliate Marketing	55.0%	£0.00	60.0%	£0.00	65.0%	£0.00
Weighted Sum	£116.10		£214.18		£127.29	

9.5.5 Business Lifetime Value (BLTV)

The (BLTV) is calculated by combining the profits from trade shows and affiliate marketing while accounting for data hosting costs. Trade show profits are derived from the revenue per business minus the expenditure per business added to the profit from affiliate marketing per business. These are then adjusted to their respective weighting for that year. As we sell our user data is sold in sizes of 20,000, we subtract the data hosting costs for the standard and premium user proportions within each year per 20,000 user block, ensuring an accurate breakdown of total business LTV.



Figure 9.5.5.1: BLTV Calculation Breakdown

Our (BAC) starts relatively low in Year 1 as we focus on targeting one large-capacity trade show, meeting our threshold of 20,000 users to begin selling data. In Year 2, we expand our strategy to include three trade shows of varying sizes, which increases customer acquisition costs but also broadens our reach. By Year 3, we target the same trade shows with increased expenditure, enabling us to **acquire 10.7% more businesses compared to Year 2**. This fluctuation in BAC translates into a progressive improvement in the BAC:BLTV ratio, culminating in an approximate **1:30** ratio by **Year 3**.



Figure 9.5.5.2: Annual BLTV and BAC:BLTV Breakdown

9.6 Summary of P&L

Our business breaks even by the end of Year 2, after which we invest **£7.6 million** to acquire a banking license. This unlocks revenue growth through interest accrual, enabling rapid expansion and positioning us for long-term success.



Figure 9.6: Cardo projected cumulative Profit and Loss for the next 3 years

A good EBITDA margin is said to be between 15-25% [23]. Year 2 and 3 show the strength of our business model with a growing EBITDA margin of 30.32%.

	Setup	Year 1	Year 2	Year 3
Revenue	234129	2,262,410	7,263,830	18,348,022
Cost of Goods	-308,476	-1,620,610	-4,685,805	-16,049,111
Gross Profit	-74,347	641,489	2,578,460	2,298,450
Gross Margin	-31.75%	28.35%	35.50%	12.53%
Sales & Marketing	-	-51,634.00	-113,900	-181,900.00
General & Admin	-1,850	-22,200.00	-22,200.00	-22,200
Office Costs	-	-69,727	-108,252	-69,727.20
Employees	-109,593	-332,397	-512,936	-714,056.68
AWS	-60,000	-207,822	-410,637	-752,165
EBITDA	-185,791	-42,290	1,410,534	558,401
EBITDA Margin	-65.51%	-10.00%	30.32%	53.49%
Corporate Tax	-	-	352,633	139.600
Net Income	-185791	-130,338	625,690	-2,041,642
Cumulative	-185,791	-316,179	309,511	-1,732,131

10 Appendix

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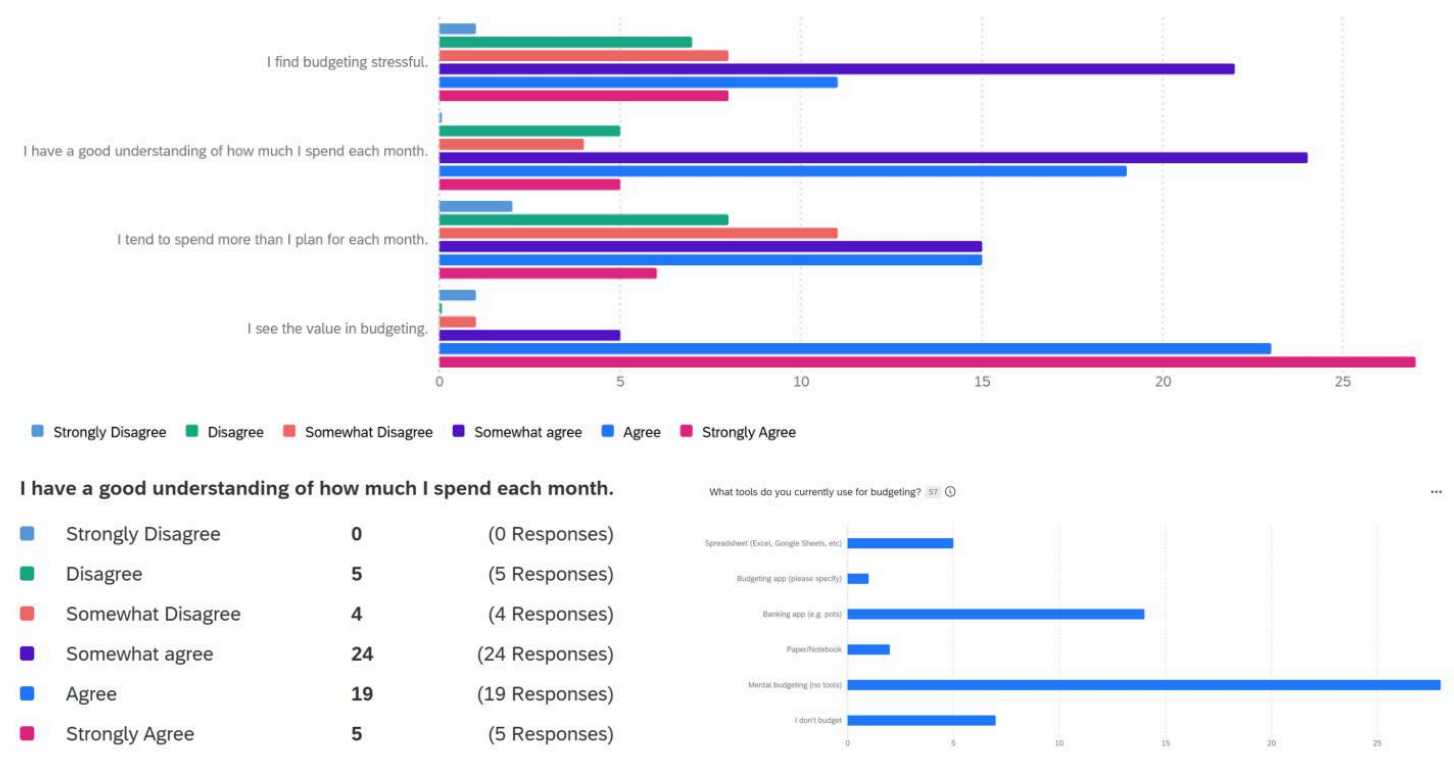
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SEE SPREADSHEET NOTES FOR FURTHER PROFIT AND LOSS REFERENCES

10 Appendix

10.1 Survey Results 1



10.2 Survey Results 2 https://qualtricsxmfn3tw6v8b.qualtrics.com/jfe/form/SV_0jgLYS4bLCX5kcS

Do you have any other thoughts about the card? 31 ⓘ

Do you have any other thoughts about the card?

Sounds a bit daft, how will colour change my focus or incentive me?

Nice idea, but I really only use Apple Pay. Also im confused if it's a bank as well or just connects to your current banking

What if i pay via apple pay? The card should show in the colour in my phone 'wallet' too

Is it gonna require network connection to constantly be updating.

It would be good because people usually use contactless without a second thought about whether they should be spending the money or not

I use apple pay more than a physical card, so a digital solution would be interesting as well

Would it be only be associated with one (current) account? Also, most people pay on their phone so would there be a way to digitise this?

I don't want others to see this information

I definitely would not want to charge it or anything like that

I think this is a new and clever way which can be very successful and helpful in aiding and guiding people on their spending. Seeing a physical warning/alert on your card would be revolutionary

Cool idea but I don't use my card anymore I just use Apple Pay

It looks cool

Different colours maybe

I think overall it's nice. But part of me wanders whether this should just be over daily budget because you would expect the card (because most people expect their budget to be like spent by the end of the month). Maybe a progressively appearing logo in regards to food, (eating out/ entertainment/travel) would be the most helpful and the logo being completely filled when your budget is up.

Where would you see this? If I have to open my banking app I can already see my remaining balance anyway and you can't do this on a physical card. It would be good to put it directly when you're about to pay with Apple / Google Pay

I would want more details as opposed to just having it as a blanket statement (eg I bought a phone that would obviously put my standard budget out, but because I'd taken more money from my savings account to do it)

I think most people just use apple pay these days. Users may feel embarrassed to use card if they are off budget. Im curious what you would make it out of. If its a screen, how will it be powered? Will i have to charge my card?

I like the idea of it changing colour to discourage spending when you can see you've overspent/red but I wonder if there is some psychology with the colours. Like if the card is green maybe you're more likely to feel ok about frivolous expenses and then always end up in like orange when actually it would be best for the card to stay green? So I guess green means "go" in our heads, when actually for this case it shouldn't

I dont often use my physical card because of Apple Pay, so would be nice if this colour change is reflected on my digital card from my apple wallet

Do you have any other thoughts about the banking app for Cardo? 31 ⓘ

...

Do you have any other thoughts about the banking app for Cardo?

I think for me I don't want to flip through day to day expenses I'd rather see more all at once and trends which is why insights is nice. Categories are helpful

The background is a good idea but is a lot of noise and very distracting. The middle page also has too much going on hard to see the clear purpose of it.

It feels like it would provide a very similar functionality that I can already get with Monzo

Monzo already does most of this

The your spending page seems like a worse implementation of the insights page. The ui is clunky and it doesn't really show any more info. I'd consider combining the two into something much cleaner

It would be helpful if the landing page noted how much was left of the budget

Do you have any overall thoughts about the idea of Cardo? 31 ⓘ

Do you have any overall thoughts about the idea of Cardo?

I would be interested to see how you'd prevent more established competitors from implementing the colour changing card and stealing your USP, as there may be different methods of colour changing technologies that can't be patented. Also maybe you can implement notifications that pop up just before payment if you are using NFC or online banking, to give users a choice on whether what they are buying really is relevant (e.g. "you are making a food payment of £X which will leave you £Y for the rest of the month. Are you sure you want to proceed?"). This may work for me because most of my impulsive spending is on food or clothes.

I think you could really benefit from having a more specific user group than just graduates and tailoring the app to them. E.g. graduates from outside London moving to the city may struggle with budgeting for different reasons than others who maybe suddenly earn a very high salary and don't know how to evaluate what a good budget is. Also is the goal to help them reduce spending (which again may not be a super high priority for grads who just earned money) or to educate on budgeting. It seems like the premium version is about reducing spending but then the overall physical idea is about visually showing your budget

Good concept but would suggest making the app your base feature and improving that and having the card as something for your premium users.

Interesting concept to help people budget, but I wouldn't spend any additional money on it than the free functionalities I already get with my banking app. The card changing colour may be a good visual indicator, but can be embarrassing for the user if other people know why it is turning red

I believe a £3.99 starting premium price would be smart as it would be a newly introduced app, where there would be great opportunities to later introduce higher memberships which should include perks & more offers or discounts through partnerships with different brands

spending per category

I think directly combining with areas where you can save money e.g. on your TFL travel charge it tells you about linking your railcard. There's loads of discounts and cash back and things but they are all very separate and you have to manually know

The spending page feels similar to Monzo's one

The premium model sounds like it doesn't add anything that free sites like moneysavingexpert give for free. If a user is consciously budgeting, they would use this free alternative over paying for yours. I think overall it's a neat but the app doesn't offer anything that other banking apps don't already offer (refer to monzo, chase, revolut). It would be worth trying to find a less explored niche - I liked the idea around IoT and physical integrations, but I don't think the card is the way to go due to social factors and it becoming less common to use. Possibly some kind of artwork in the house that reflects monthly budget in a subtle way? E.g. a model flower that blooms when on budget or withers when off budget.

10.3 Market Opportunity Screenshots

Estimating data insights value

The value of your data to the online advertising industry can be estimated to be around \$263 per year[<https://www.invisibly.com/learn-blog/how-much-is-data-worth/>]. Say we capture 50% of a users spending data, say spending data contributes 5 to 25% of a users data value. that would results in our RPU being between **£10.33 GBP to £51.66 GBP**.

Estimating interest acrural insights value

Assuming we can capture 90% of this for interest gains, that would result in 491 gbp per user. Assuming a 5% interest rate environment that could result in 24\$ ARPU on interest gains. Based on wealth levels across different age ranges we can assume [<https://www.businessofapps.com/data/revolut-statistics/>] the older one is the more money they will have on their account. This corresponds to 390 gbp for young people, 545 for middle aged and 700 for old people.

Estimating subscription value

Premium subscription will add avg of 2.5\$ per user. from our survey 31% of people would pay for the premium subscription. therefore $0.32 * (2.5 * 12) = 9.6$ per user

Estimating interchange fees

Interchange fees for transactions involving UK-issued cards are limited to 0.3% of the total for credit card payments and 0.2% for debit card payments. Interchange fees for cross-border transactions can be up to 1.5%. [<https://www.barclaycard.co.uk/content/dam/barclaycard/documents/business/help-and-support/interchange-rates-and-scheme-fee-guide.pdf>]

Estimating advertisement fees

Total Daily Sessions:

$10,000 \text{ users} \times 3 \text{ sessions/user} = 30,000 \text{ sessions}$

Total Daily Minutes:

$30,000 \text{ sessions} \times 5 \text{ minutes/session} = 150,000 \text{ minutes}$

Total Daily Ad Impressions:

$150,000 \text{ minutes} \times 2 \text{ ads/minute} = 300,000 \text{ impressions}$

Adjusted for Fill Rate:

$300,000 \text{ impressions} \times 90\% = 270,000 \text{ filled impressions}$

Daily Revenue:

$(270,000 \text{ impressions} \div 1,000) \times £2 \text{ eCPM} = £540$
Monthly Revenue:

$£540/\text{day} \times 30 \text{ days} = £16,200$

Annual Revenue:

$£16,200/\text{month} \times 12 \text{ months} = £194,400$

ARPU:

$£194,400 \text{ annual revenue} \div 10,000 \text{ users} = £19.44$
per user per year

Benchmarking other brands ARPU

Revolut = 36\$

Monzo = 90\$

Wise = 85\$

Sensitivity analysis

	BAD	EXPECTED	GOOD	BAD	EXPECTED	GOOD	BAD	EXPECTED	GOOD
PEOPLE		5.3			3.5			2.53	
%	1.5	2.25	3	1.5	2.25	3	1.5	2.25	3
ADS	15	19.44	25	15	19.44	25	15	19.44	25
INTEREST	1.17	13.65	39	1.635	19.075	54.5	2.1	24.5	70
SUBS	7	9.6	12	7	9.6	12	7	9.6	12
DATA	10.33	30.995	51.66	10.33	30.995	51.66	10.33	30.995	51.66
INTERCHANGE	0.234	0.546	1.17	0.327	0.763	1.635	0.42	0.98	2.1
	2681853	8852046.75	20483970	1800330	6289998.8	15203475	1322557.5	4867941.38	12201684
	5604740.5	20009986.9	47889129						
	5.8047405	20.0099869	47.889129						

10.4 RICE Framework

(

Reach

x

Impact

x

Confidence

)

/

Effort

Sales Channel	Reach	Impact	Confidence	Effort	RICE Score
Transport Ads	9	8	80%	5	11.52
Social Media Ads	7	8	80%	5	8.96
Trade Shows	5	7	70%	6	4.67
Webinars	3	8	80%	7	2.74
SEO	6	3	60%	5	2.88
Email Marketing	3	4	80%	4	4.80
Unconventional Pr	8	9	20%	8	1.80

10.4 Profit and Loss Excel Spreadsheet

[See full P&L spreadsheet here](#)

References are also available are on the P&L